



Deusto

Facultad de Ingeniería
Ingeniaritza Fakultatea

Grado en Ciencia de Datos e Inteligencia Artificial

Datuen Zientzia eta Adimen Artifizialeko gradua

Proyecto fin de grado

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Exploring daily emotional well-being through
personalized data analysis

Adriana Rodríguez Rodríguez

Director: Alexandre Barco Martelo

San Sebastián, mayo de 2026



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Abstract

This project analysed daily emotional well-being from a personalized perspective using records of emotional states and contextual information derived from daily life. It was based on the idea that well-being is not a fixed value, but varies from day to day, and can be better understood in relation to individual context and behaviour. The work is framed within the field of Data Science and focuses on emotional states at an individual level.

The project focused on constructing an integrated daily dataset from multiple personal data sources and analysing emotional states. The distributions of emotional variables, their temporal evolution and the differences between users were studied, together with their variability over time.

Personality and well-being questionnaires were also incorporated as stable references to better contextualize daily emotional results. A simple personalized recommendation approach was then developed by comparing similar days within each user, exploring how past contextual conditions relate to differences in the user's emotional state.

Finally, machine learning techniques were used to evaluate whether daily contextual information contains signals that help distinguish between different types of days relative to a person's usual well-being. The project adopted an applied approach, with an exploratory scope focused on understanding patterns rather than building a fully predictive system.

The results showed that emotional well-being is very personal and changes from day to day. General models had limited value, while personalized approaches captured these differences much better, highlighting the importance of analysing each person on their own terms.

Keywords

Personalized well-being analysis
Daily contextual data
Emotional data analysis
Multimodal data

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1 INTRODUCTION

1.1 Background and Context

In recent years, the study of human behaviour and mental health has gained increasing relevance, partly due to the availability of digital data collected through personal devices such as smartphones [1]. Research on digital phenotyping (i.e., behavioural data collected via personal digital devices) and well-being increasingly combines information from active inputs such as questionnaires and self-reports with passive sensing signals such as mobile phone usage, physical activity, mobility, and sleep-related indicators [2].

These data are typically collected in a longitudinal manner, which makes it possible to observe how emotional and behavioural states evolve over time [3]. Questionnaires and self-reports provide subjective information about psychological state, while sensing measures offer more objective indicators of daily behaviour. Recent literature increasingly adopts multimodal approaches that combine self-report measures and passive sensing data to study behaviour and well-being [2].

Working with this type of information involves handling data coming from different formats and measurement levels, which usually requires prior preparation, organization, and integration steps before analysis can be carried out. For this reason, studies in this area typically include a dedicated data construction and structuring phase.

This work uses the StudentLife Dataset [4], a longitudinal dataset collected in a university setting that combines psychological questionnaires, daily self-reports, and multiple mobile sensing signals. This type of dataset makes it possible to jointly study behaviour, context, and emotional states under real-life conditions.

1.2 Problem Statement

Analysing emotional well-being using digital data is not straightforward, even when rich datasets are available. The information usually comes from different sources such as questionnaires, daily self-reports, and sensor-based measurements, and these sources do not share the same format or structure. As a result, the data must be organized and integrated before it can be used for meaningful analysis.

In addition, real-world datasets often contain missing or incomplete values. Not all variables are recorded with the same regularity: some are measured many times per day, while others are collected only occasionally. This makes direct comparison difficult and requires careful preparation at a common daily level.

For these reasons, there is a need for analysis approaches that work with properly structured daily data and that consider behavioural and emotional information together rather than in isolation.

1.3 Motivation

The motivation for this work comes from the interest in better understanding how daily collected data from multiple sources can be used to analyse emotional well-being in real-life contexts. By combining different types of data, it becomes possible to obtain a richer and more comprehensive view of individuals' everyday experiences. Unlike single-point measurements, continuous records make it possible to observe how emotional and behavioural states evolve over time.

Working with this type of information is particularly appealing because it combines emotional and behavioural dimensions, allowing a more complete view of a person's day-to-day activity. In addition, longitudinal data makes it possible to observe how records change over time rather than relying on a single isolated measurement.

Within this context, the project is framed as an applied exploration of how to structure, analyse, and make use of this type of data in order to obtain useful and understandable insights from daily records. To better position this work within the existing research on the StudentLife dataset, it is useful to briefly review how previous studies have approached the analysis of this data.

1.4 Relation to Previous StudentLife Research

Several relevant studies based on the StudentLife dataset have focused on different aspects, including dataset presentation, analysing behavioural correlations, and developing predictive models for specific mental-health-related outcomes [4] [5] [6]. The original StudentLife paper introduced the StudentLife sensing system and dataset and explored correlations between automatically sensed smartphone behavioural data and students' academic performance and mental well-being measures over an academic term [4].

Later works shifted the focus toward more specific predictive tasks, such as stress prediction using smartphone sensing data and personalized stress prediction using deep learning models [5] [6] .

In contrast to previous works that mainly focus on specific prediction tasks, the present project explores a broader analytical perspective on daily emotional well-being using the StudentLife dataset. Instead of concentrating exclusively on predictive modelling, the work adopts a broader perspective that combines behavioural analysis and interpretable modelling approaches to better understand daily emotional well-being from behavioural and self-reported data.

This perspective motivates the development of the present study and leads to the set of objectives described in the following section.

1.5 Objectives

The main objective of this project is to analyse daily emotional well-being using longitudinal multimodal behavioural and self-report data through a structured and interpretable analytical approach.

To achieve this goal, the specific objectives are:

- To prepare and integrate daily data from questionnaires, self-reports, and behavioural records into a consistent analytical dataset.
- To perform a structured exploratory analysis of daily emotional data using descriptive statistics and visualizations.
- To measure emotional variability and stability at the individual level.
- To derive interpretable well-being and personality-related profiles from survey data.
- To explore personalized context-based recommendation strategies for daily emotional well-being.
- To analyse whether contextual and behavioural factors are associated with daily emotional well-being.
- To reflect on the ethical implications of working with personal data.

1.6 Scope of the Project

This project focuses on the applied analysis of daily emotional well-being data using a longitudinal multimodal dataset collected in a university context. The scope of the work is limited to the processing, structuring, and analytical study of the available records, as well as the development of interpretable modelling and recommendation approaches to extract meaningful insights from the data.

The project is developed as an analytical and research-oriented study and does not aim to produce a system or an operational deployment. The techniques used are intended for data exploration and pattern understanding rather than for clinical assessment or psychological decision-making.

The conclusions obtained refer only to the analysed dataset and the specific study context. No direct generalisation to other populations or environments is claimed without further validation.

1.7 Project Planning

Every project is generally developed through five main stages: definition, planning, execution, control, and completion. In this project, the planning phase followed an iterative and progressive methodology, allowing the objectives, tasks, and development phases to evolve throughout the project lifecycle.

The initial idea of the project emerged in May 2025, when the StudentLife dataset was identified as the basis for studying emotional well-being. During this first stage, the initial objectives of the project were established, focusing on the analysis of emotional well-being patterns and the integration of heterogeneous data sources. As the project progressed and a deeper understanding of the dataset was achieved, additional objectives and analytical approaches were progressively incorporated.

The planning phase involved dividing the project into its main objectives and development tasks, organizing the sequence of activities, estimating the required time, and identifying the main milestones. The project was structured into several stages, including data preparation and cleaning, analytical dataset construction, exploratory analysis, emotional variability studies, profile

characterization, recommendation system development, machine learning analysis, and final documentation. Due to the iterative methodology, the planning was continuously revised and adapted throughout the project.

The technical development started in May 2025 with the data preparation and cleaning phase (Chapter 2), which was the most time-consuming stage due to the complexity and heterogeneity of the dataset. This phase included temporal normalization, treatment of missing values, integration of heterogeneous sources, and analytical dataset construction.

Between July and November 2025, the project was temporarily paused. The work resumed in December 2025 with the validation of the analytical dataset and the revision of the preprocessing pipeline before continuing with the analytical stages.

From January 2026 onwards, the project advanced through different analytical phases, including exploratory emotional analysis (Chapter 3), emotional variability and stability studies (Chapter 4), survey-based personality and well-being profiles (Chapter 5), the development of a similarity-based recommendation system (Chapter 6), and machine learning experimentation to identify factors associated with emotional well-being (Chapter 7).

The final stage focused on the interpretation of results, ethical considerations, final report writing, revision, formatting, and preparation of the final submission. The dissertation was written progressively throughout the project, although the final stages required additional work related to revision and cohesion.

The main milestones of the project included the completion of the data preparation phase, the validation of the analytical dataset, the completion of each analytical chapter, and the final submission of the dissertation. Continuous monitoring and coordination with the supervisor were maintained throughout the project.

Figure 1 summarizes the overall project schedule through a Gantt chart representing the main development phases, milestones, estimated workload, and temporal distribution between May 2025 and May 2026.

Gantt Chart — Project Planning

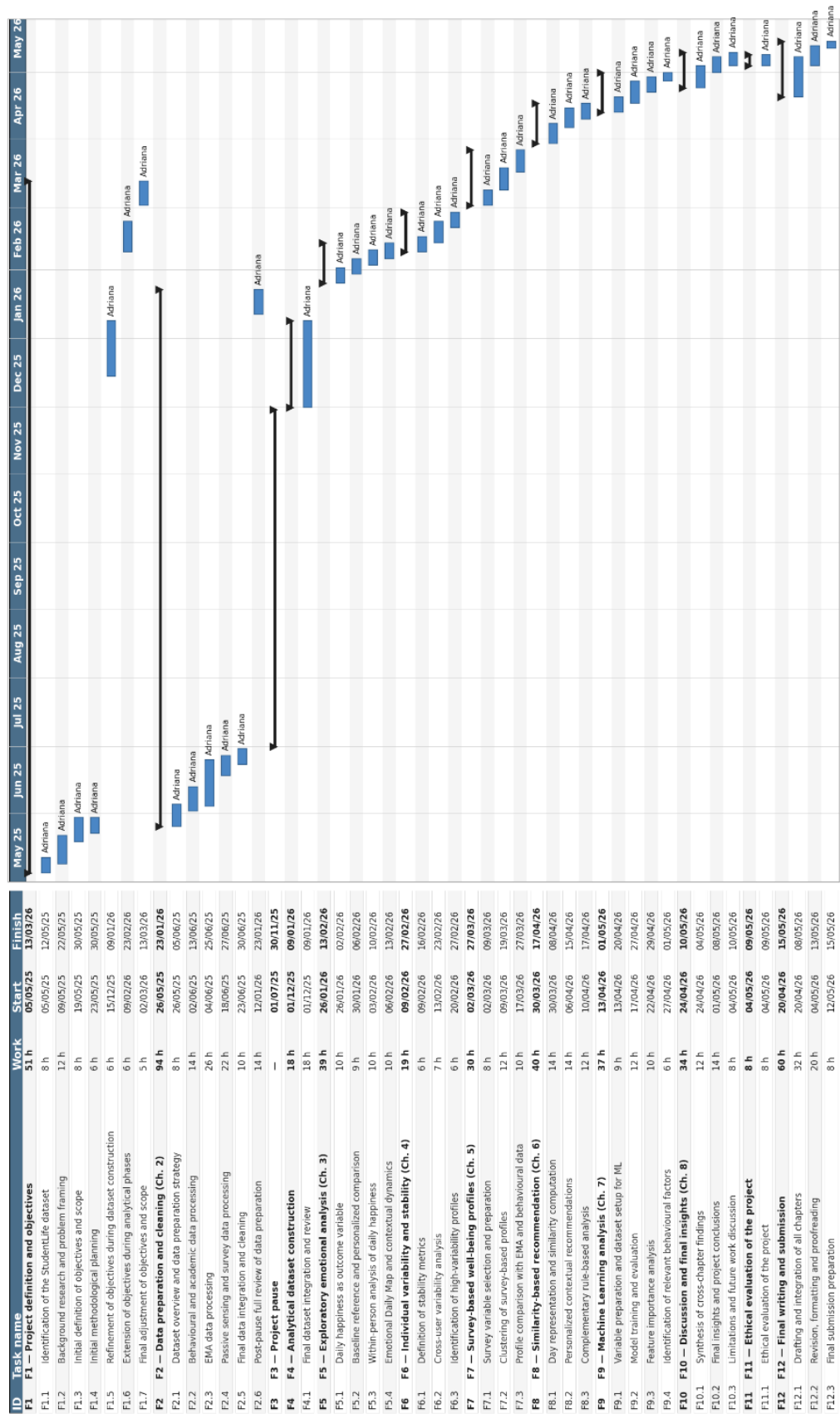


Figure 1. Gantt Chart of the Project Planning

1.8 Budget and Project Cost Evaluation

The economic cost of the project has been estimated based on the working hours invested in each task, valued at 20 € per hour. The detailed breakdown by phase and task is shown in Table 2.

1.8.1 Personnel Costs

Personnel costs correspond to the dedication throughout the different phases of the project, from the initial definition of objectives to the final submission of the dissertation. The total estimated dedication is 430 hours, which corresponds to a personnel cost of 8,600 €.

1.8.2 Other Costs and Total Project Cost

Table 1. Project Budget Breakdown by Phase and Task

ID	Phase / Task	Hours	Cost (€)
F1	F1 — Project definition and objectives	51	1020
F1.1	Identification of the StudentLife dataset	8	160
F1.2	Background research and problem framing	12	240
F1.3	Initial definition of objectives and scope	8	160
F1.4	Initial methodological planning	6	120
F1.5	Refinement of objectives during dataset construction	6	120
F1.6	Extension of objectives during analytical phases	6	120
F1.7	Final adjustment of objectives and scope	5	100
F2	F2 — Data preparation and cleaning (Ch. 2)	94	1880
F2.1	Dataset overview and data preparation strategy	8	160
F2.2	Behavioural and academic data processing	14	280
F2.3	EMA data processing	26	520
F2.4	Passive sensing and survey data processing	22	440
F2.5	Final data integration and cleaning	10	200
F2.6	Post-pause full review of data preparation	14	280
F3	F3 — Project pause	—	—

ID	Phase / Task	Hours	Cost (€)
F4	F4 — Analytical dataset construction	18	360
F4.1	Final dataset integration and review	18	360
F5	F5 — Exploratory emotional analysis (Ch. 3)	39	780
F5.1	Daily happiness as outcome variable	10	200
F5.2	Baseline reference and personalized comparison	9	180
F5.3	Within-person analysis of daily happiness	10	200
F5.4	Emotional Daily Map and contextual dynamics	10	200
F6	F6 — Individual variability and stability (Ch. 4)	19	380
F6.1	Definition of stability metrics	6	120
F6.2	Cross-user variability analysis	7	140
F6.3	Identification of high-variability profiles	6	120
F7	F7 — Survey-based well-being profiles (Ch. 5)	30	600
F7.1	Survey variable selection and preparation	8	160
F7.2	Clustering of survey-based profiles	12	240
F7.3	Profile comparison with EMA and behavioural data	10	200
F8	F8 — Similarity-based recommendation (Ch. 6)	40	800
F8.1	Day representation and similarity computation	14	280
F8.2	Personalized contextual recommendations	14	280
F8.3	Complementary rule-based analysis	12	240
F9	F9 — Machine Learning analysis (Ch. 7)	37	740
F9.1	Variable preparation and dataset setup for ML	9	180
F9.2	Model training and evaluation	12	240
F9.3	Feature importance analysis	10	200
F9.4	Identification of relevant behavioural factors	6	120

ID	Phase / Task	Hours	Cost (€)
F10	F10 — Discussion and final insights (Ch. 8)	34	680
F10.1	Synthesis of cross-chapter findings	12	240
F10.2	Final insights and project conclusions	14	280
F10.3	Limitations and future work discussion	8	160
F11	F11 — Ethical evaluation of the project	8	160
F11.1	Ethical evaluation of the project	8	160
F12	F12 — Final writing and submission	60	1200
F12.1	Drafting and integration of all chapters	32	640
F12.2	Revision, formatting and proofreading	20	400
F12.3	Final submission preparation	8	160
	TOTAL	430	8,600

1.9 Methodological Overview

The project follows a phased analytical workflow based on a longitudinal multimodal dataset. The process starts with dataset understanding, selection of relevant sources, and construction of a daily-level analytical table through data preparation, aggregation, and integration steps.

Once the working dataset is built, an exploratory analysis phase is conducted to characterize emotional records through descriptive measures and visualizations. This stage supports the identification of general distributions, temporal behaviour, and basic patterns in the data.

Next, individual-level analyses are performed to study emotional variability and stability across time. In a complementary step, survey information is used to obtain interpretable well-being and personality-related profiles.

Based on the daily records, context-based recommendation approaches are applied to analyse behavioural patterns and derive personalized insights.

A modelling analysis is conducted to examine whether contextual and behavioural factors are associated with daily emotional well-being.

The results obtained across the different stages are then brought together in a final discussion that synthesizes the main findings, presents the overall insights of the project, and identifies its main limitations and future directions.

Finally, the workflow is complemented by an ethical reflection on the implications of working with sensitive personal data.

Figure 3 summarizes the methodological framework of the project, illustrating the main analytical stages and their relationships.

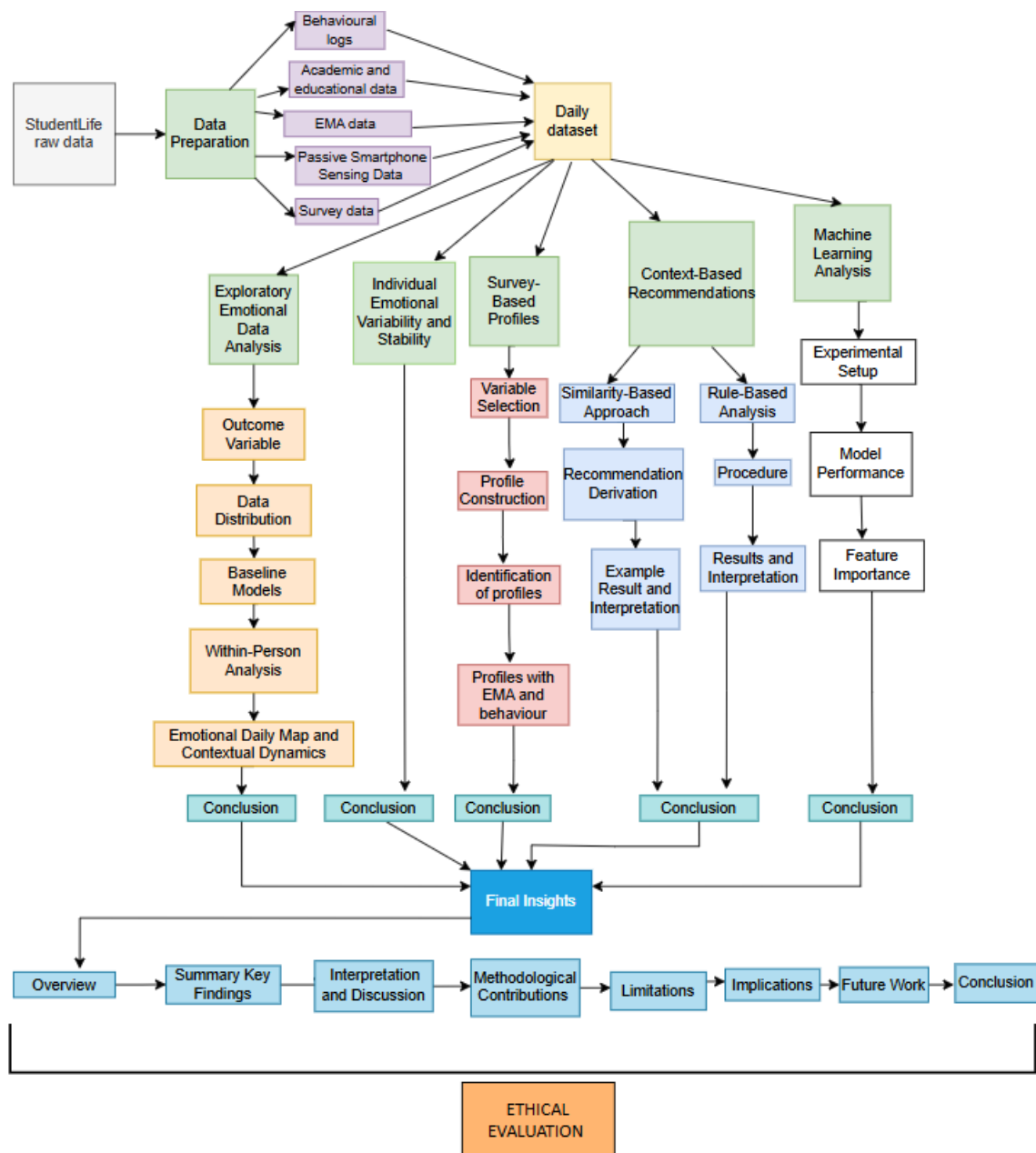


Figure 1. Methodological Framework of the Project

1.10 Document Structure

The structure of this document is organized as follows. Chapter 2 describes the dataset and the data preparation process. Chapter 3 presents the exploratory emotional analysis and descriptive results. Chapter 4 focuses on individual variability and stability analysis. Chapter 5 introduces the survey-based profile construction. Chapter 6 presents context-based recommendation approaches for personalized well-being insights. Chapter 7 examines how contextual and behavioural factors relate to daily emotional well-being. Chapter 8 presents the discussion of the results, the final insights of the project, and possible future work. Finally, Chapter 9 develops the ethical evaluation of the project, addressing the normative and transversal aspects of working with this kind of data.

2 DATA PREPARATION AND CLEANING

This chapter describes the data preparation and integration process used in the project. Given the complexity of the dataset, a structured pipeline was defined to organize the different stages of preprocessing.

The overall structure of this process is shown in Figure 2.

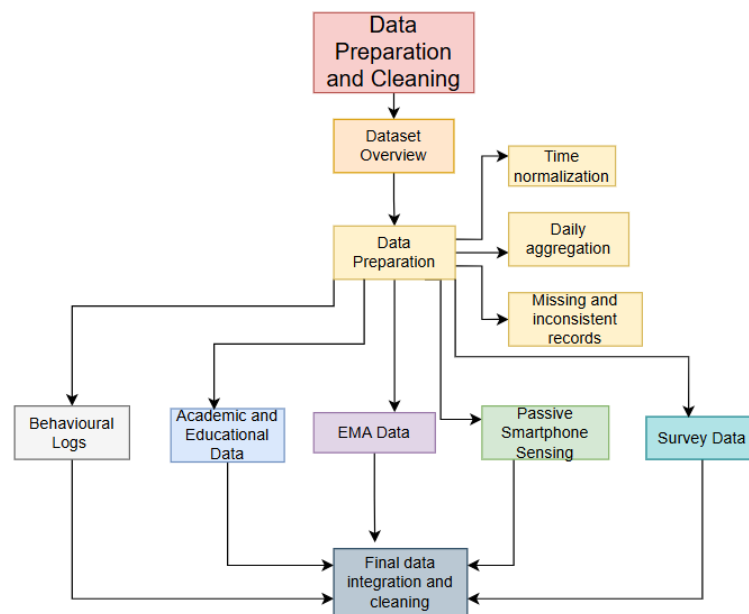


Figure 2. Data Preparation and Integration Pipeline

2.1 Dataset Overview

For this project, the StudentLife dataset was used. This dataset was collected as part of a longitudinal study conducted at Dartmouth College, where university students were monitored throughout an academic term [4]. The goal of the study was to capture daily behavioural and psychological patterns in real-world conditions.

The dataset combines multiple types of information, including passive smartphone sensing, behavioural and academic logs, ecological momentary assessment (EMA) questionnaires, and validated psychological surveys. These sources provide complementary views of daily behaviour, context, and emotional state.

Because the data are collected repeatedly over time and across different domains of daily life, the dataset enables the study of emotional well-being from a longitudinal and contextual perspective. This makes it suitable for applied analytical work focused on daily emotional patterns and individual-level variation.

The dataset originates from the StudentLife study, a research project conducted at Dartmouth College during the Spring term of 2013, in which students from a computer science course voluntarily participated by installing a smartphone application that continuously collected behavioural sensing data. The study initially involved 60 students. They were monitored throughout the academic term using their smartphones. Alongside this automatic data collection, participants also completed ecological momentary assessments (EMAs), which consist of real-time self-reports collected during their daily life, as well as several validated psychological surveys administered at the beginning and end of the study period.

A key aspect of the study is the use of smartphones as a continuous sensing platform, allowing behavioural data to be collected automatically throughout the day while participants go about their normal activities.

The dataset was introduced in the original StudentLife paper [4], which presents a continuous smartphone sensing system used to assess the impact of workload on students' stress, sleep, activity, mood, sociability, and mental well-being during an academic term at Dartmouth College. In the analyses reported in the paper, 48 students completed the study and were included in the final analysis. Among these participants, 30 were undergraduate students and 18 were graduate students. The paper also reports that 10 participants were female and 38 were male, and that the sample included 23 Caucasian students, 23 Asian students, and 2 African-American students. Although several participant characteristics are described, the individual ages of the students are not explicitly reported. However, the publicly released dataset contains additional participant identifiers, and the number of users with available observations may vary depending on preprocessing and data aggregation steps.

2.2 Data Preparation Strategy

Before conducting the analysis, several preprocessing steps were applied to the raw StudentLife dataset to ensure temporal consistency, structural alignment between data sources, and reliable analytical units. The preparation process involved three main stages: time normalization and timestamp handling, aggregation of event-level records into daily observations, and the treatment of missing or inconsistent values. These stages were applied sequentially to progressively transform heterogeneous event-level records into a consistent daily analytical dataset.

2.2.1 Time Handling and Time Zone Normalization

In the raw StudentLife dataset, most behavioural records are timestamped using UNIX epoch time in UTC (e.g., 1364437400). To enable consistent daily analysis, all timestamps were converted into a standard datetime format and adjusted to the America/New York time zone, which corresponds to the geographical context of the study. These timestamps were then transformed into a daily date format (e.g., 2013-03-27) to ensure consistent aggregation.

This conversion ensures that daily aggregates correctly reflect participants' real-life routines and prevents records from being assigned to incorrect calendar days due to time zone offsets. After normalization, a local calendar date variable was derived from each timestamp.

This date field was then used as a core temporal reference throughout the project and later combined with the user identifier to define the daily analysis unit.

```
"resp_time": 1364437400
```

Figure 3. Example of a Raw UNIX Epoch Timestamp (`resp_time`) from the StudentLife EMA Dataset Before Temporal Normalization

```
2,2013-03-27,2
```

Figure 4. Conversion of UNIX Epoch Timestamp to Local Datetime

2.2.2 Daily Aggregation and Analysis Unit

Since the different dataset sources contain event-level records (such as app usage logs, device interactions, and momentary questionnaire responses), it was necessary to define a common unit of analysis to integrate all information in a consistent way.

In this project, the unit of analysis was defined at the user–day level, using (`user_id`, `date`) as the primary key. This makes it possible to aggregate all records into a consistent daily format, enabling comparison across data sources and analysis of daily behavioural patterns.

Based on this definition, the data were aggregated by user and by day using count, sum, mean, and other descriptive operations depending on the variable type. As a result, each row in the final analytical dataset represents a daily summary of behavioural and self-report information for each participant.

Working at the daily level also enables temporal comparisons over time, the analysis of individual variability, and the integration of behavioural and questionnaire-derived measures within a unified analytical framework.

```
user_id,date,
0,2013-03-23,
0,2013-03-24,
0,2013-03-25,
```

Figure 5. Example of the User-Day structure after temporal aggregation, where each row represents one user on a specific day

2.2.3 Handling Missing and Inconsistent Records

Given the heterogeneous origin of the dataset and the different collection mechanisms involved, missing and inconsistent records were expected and explicitly addressed during preprocessing.

First, invalid or corrupted values detected during file parsing and variable extraction were coerced to missing values and excluded when they could not be reliably interpreted. Records were also examined

for inconsistencies in timestamps or identifiers as part of a validation step to ensure data consistency. When such issues were detected, they were removed to prevent potential alignment errors in later aggregation steps (e.g., non-numeric timestamps coerced to NaN and discarded, missing temporal fields such as `resp_time`, or invalid intervals where end times preceded start times; although no critical inconsistencies were ultimately detected, these rules were included as a precautionary validation step).

A distinction was made between true absence of activity and absence of measurement. For behavioural variables expressed as counts, durations, or event totals, missing values were replaced with zero when the absence clearly indicated that no activity occurred on that day. This approach was applied to variables such as application usage counts, calendar events, call activity, meal records, academic deadlines, EMA participation indicators (e.g., comment and response counts), and passive sensing indicators (e.g., Bluetooth scans, screen inactivity, WiFi observations, and SMS activity), where missing values can be reasonably interpreted as absence of activity. Additionally, certain user-level indicators, such as the number of enrolled courses (`num_classes`), were also assigned a value of zero when missing, under the assumption that the absence of enrolment records reflects no registered courses for that participant.

In contrast, when a variable was not measured or no response was available, particularly in questionnaire-based data, the value was kept as missing (NaN). This distinction preserves the semantic difference between “zero activity” and “no information available,” which is important for correct interpretation in subsequent analyses.

To facilitate readability, the data processing pipeline is organized into several sections according to the nature of the data sources. First, behavioural logs are described, followed by academic and educational data, EMA self-reported measures, passive smartphone sensing data, and survey-based variables. This structure allows a clear distinction between objective behavioural records, contextual academic information, subjective self-reports, and baseline individual characteristics.

2.3 Behavioural Logs Processing

In this study, behavioural logs are broadly defined as objective records capturing observable aspects of participants’ daily behaviour and routines. This includes direct interaction data (e.g., app usage and call activity) as well as structured records reflecting daily organization and habits (e.g., calendar events and meal patterns). After defining the general preprocessing strategy, the following sections describe how the main behavioural data sources were processed and aggregated at the daily level.

2.3.1 App Usage

App usage records were aggregated at the daily level for each participant in order to obtain a compact and comparable representation of mobile phone interaction patterns. From the processed usage logs, several summary indicators were computed per user and day.

The variable *total_apps_used* represents the number of different applications used during the day and reflects the diversity of app usage. The variable *most_used_app* identifies the application with the

highest number of recorded interactions on that day, providing a simple indicator of dominant usage behaviour.

The indicator *num_sessions* corresponds to the total number of recorded usage events and reflects how frequently the user interacts with the device.

In addition, *tiempo_uso_total* represents the total amount of activity recorded across the day, computed as the sum of activity counts. Although its name may suggest a time measure, it does not represent actual usage time, but rather the overall level of activity generated by the device.

For example, applications such as a web browser usually generate higher activity values because the user interacts more with them (loading pages, navigating between screens), while simpler system interfaces such as the home screen typically produce lower values. Together, these daily indicators provide a structured summary of application usage behaviour and are later used as behavioural features in the exploratory and analytical stages of the project.

2.3.2 Calendar

Calendar records were used to derive a daily indicator of scheduling activity. Each file contains timestamped event entries associated with a specific participant.

After assigning events to calendar dates, the data were aggregated at the daily level. For each (user_id, date) pair, a single summary feature was computed:

- *calendar_event_count*: number of calendar events recorded on that day.

This feature provides a compact representation of daily planning intensity while avoiding the use of event content or semantic labels.

2.3.3 Call Log

Call log records were processed to obtain a daily measure of communication activity. Each dataset contains timestamped call events linked to an individual participant.

Records were grouped by user and date to compute a daily summary feature:

- *total_calls*: number of call events recorded during the day, including incoming, outgoing, and missed calls.

Only event counts were retained, excluding additional call attributes, in order to produce a stable and privacy-preserving indicator of communication behavior.

2.3.4 Dining

Dining records were transformed into daily features representing meal occurrence patterns. Each entry corresponds to a registered meal event associated with a participant and a calendar date.

Meals were categorized into three types: breakfast, lunch, and dinner. Based on this classification, daily summary features were computed for each (user_id, date) pair:

- *total_meals*: total number of meals recorded on that day
- *breakfast_events*: number of breakfast records
- *lunch_events*: number of lunch records
- *supper_events*: number of dinner records

These variables provide a compact representation of daily eating structure and routine without requiring detailed nutritional information.

2.4 Academic and Educational Data Processing

Educational datasets were incorporated to provide contextual academic information that complements daily behavioural and emotional records. These sources describe workload, performance, deadlines, and online course participation. Since the data come in different formats and structures, each source was processed separately through cleaning, validation, and restructuring steps. The raw data were then converted into simple indicators by counting elements (such as enrolled courses or daily deadlines), computing summary measures (such as average grades and their variability), and combining interaction records to quantify engagement. Finally, all variables were organized in a consistent format to be integrated into the final analytical dataset.

2.4.1 Enrolment and Workload

Enrolment records were used to derive an indicator of academic workload. Each row corresponded to a student and contained multiple course identifiers. After removing non-informative fields and validating subject codes, the number of enrolled courses (*num_classes*) was computed for each participant. This variable reflects the overall academic load during the term.

2.4.2 Grades and Performance Indicators

Grade data were processed to obtain simple summary indicators for each student. All grade values were converted into numeric format using a GPA scale (e.g., values such as 2.8, 3.5 or 4.0), where higher values indicate better academic performance. The overall GPA was used as the main indicator of academic performance (*grade_mean*). Additionally, two complementary grade fields were used: *gpa_13s*, which represents the GPA obtained during a specific academic term, and *cs_65*, which corresponds to the grade in a particular course. When these values were available, a variability measure (*grade_std*) was computed to capture differences between overall academic performance and specific results.

2.4.3 Academic Deadlines

Deadline data were reshaped into a date-based format to enable daily alignment with the rest of the dataset. From these records, a daily count of academic deadlines (*num_deadlines_today*) was

computed. This variable acts as a short-term academic pressure indicator and can be used as contextual information in later behavioural analyses.

2.4.4 Online Platform Engagement

Online course platform data were processed to quantify student participation in academic discussions. After standardizing user identifiers, counts of posted questions and responses were combined into a single engagement score per participant (*piazza_score*), calculated as the sum of both variables. This indicator reflects active involvement in course-related online activity and serves as an additional academic context variable. The resulting dataset included the variables *num_posts*, *num_responses*, and the aggregated engagement indicator *piazza_score*.

2.5 EMA Data Processing

This section describes how EMA data were processed to transform original responses into structured daily-level variables. Relevant questionnaires were selected and responses were aggregated for each user and day. The aim is to obtain clear and interpretable daily indicators for analysis.

2.5.1 EMA Data Selection and Processing Framework

2.5.1.1 Role of EMA Data in the Study

Ecological Momentary Assessment (EMA) data were incorporated as the self-report component of the dataset, capturing in-the-moment information about participants' daily activities and emotional states. These responses provide subjective indicators collected close to real experiences in time, which makes them especially valuable for behavioural and well-being analysis.

EMA measures were used to complement passive sensing and contextual records, adding a personal and perceptual layer that cannot be directly inferred from sensor or system-generated data alone.

2.5.1.2 EMA Instrument Selection and Exclusion Criteria

EMA questionnaires were selected based on three main criteria: relevance to the academic and well-being objectives of the study, sufficient data completeness across participants, and the ability to provide regular and comparable daily indicators.

Following these criteria, core modules such as Activity, Mood, Sleep, Stress, and Social were retained, as they capture recurring aspects of students' daily behaviour and emotional state and are consistently available across users.

In contrast, questionnaires that did not meet these conditions were excluded. This includes modules focused on isolated events or administrative contexts (e.g., Boston Bombing, Dimensions Protestors, Cancelled Classes, Administration's Response), as well as system-level interaction tools (e.g., QR_Code, PAM), since they do not reflect stable or recurring daily patterns. Modules related to specific events

or institutional initiatives (e.g., Green Key 1, Green Key 2, Dartmouth Now) were also excluded for the same reason.

Additionally, some instruments such as Exercise and Lab were not incorporated because their content overlapped with other variables already available in the dataset or because their responses were sparse and not consistently recorded across participants.

This selection ensures that the retained EMA variables provide stable, interpretable, and comparable daily indicators aligned with the objectives of the study.

Figure 6 provides a visual summary of this selection, distinguishing between the EMA modules that were retained and those that were excluded.

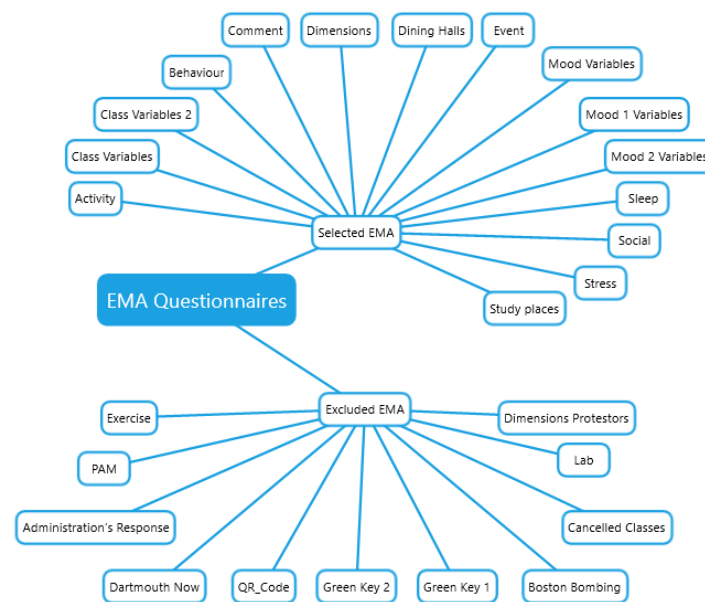


Figure 6. EMA Instrument Selection

2.5.2 Activity

The EMA Activity block captures self-reported information about how participants distributed their time across different types of daily activities. Each response reflects the perceived proportion of time dedicated to each activity category since the previous EMA prompt.

All variables in this block are measured using an ordinal five-point scale (1–5). Higher values indicate that the participant reported spending a greater proportion of time in the corresponding activity category during that interval.

The following activity variables were retained:

- *working*: proportion of time devoted to academic or work-related activities, such as studying, attending classes, or completing coursework.
- *relaxing*: proportion of time spent in leisure or relaxation activities, including resting or entertainment.
- *other_working*: proportion of time dedicated to non-academic productive tasks, such as household responsibilities or personal obligations.
- *other_relaxing*: proportion of time spent in informal or unstructured relaxation not included in the main relaxing category.
- *social2*: proportion of time spent interacting socially with other people, regardless of the specific activity performed.

For each user and day, when multiple EMA responses were available, the daily mean value of each activity variable was computed to obtain a stable daily-level representation. In days without EMA responses, values were kept as missing (NaN), since these variables correspond to self-reported measures rather than passively recorded durations, and therefore the absence of data cannot be interpreted as zero activity.

2.5.3 Class Variables 1

This EMA block captures students' subjective evaluations of specific class sessions and associated academic workload outside class time.

The following variables were processed:

- *experience*: self-reported enjoyment of the class attended that day, measured using a five-level response scale evaluating agreement with the statement "I enjoyed the class today."
- *hours*: number of hours the student reported spending on coursework outside class since the previous class session. This variable represents perceived out-of-class academic workload and was treated as a quantitative measure.
- *due*: binary indicator of academic pressure for that day, reflecting whether an assignment, quiz, or exam deadline was present (1 = yes, 0 = no).

When multiple responses were available for the same user and day, experience was aggregated using the daily mean, hours using the daily sum, and due using the daily mean to preserve the presence intensity of academic deadlines.

2.5.4 Class Variables 2

A second class-focused EMA instrument captured broader weekly academic perceptions. This block reflects students' perceived course difficulty, invested effort, and expected academic performance.

The retained variables were:

- *challenge*: response to the statement "Last week's lab was difficult and challenged me", measured on a six-point ordinal scale ranging from strong agreement (1) to strong disagreement (6), without a neutral option.

- *effort*: response to the statement “I put a great deal of effort into the course last week”, measured on the same six-point ordinal scale.
- *grade*: response to a self-reported expected course grade (e.g., A, B+, C), encoded numerically according to the questionnaire definition as an ordinal variable, where lower values correspond to higher expected performance.

When multiple responses were recorded for the same user and date, variables were aggregated using the daily mean to obtain a stable daily perception indicator.

2.5.5 Behaviour

The Behaviour EMA block captures short-term self-reported behavioral and affective states experienced by participants shortly before completing each questionnaire. Unlike baseline personality traits, these variables are designed to reflect momentary psychological and behavioural expressions, providing a fine-grained view of daily emotional and behavioural fluctuations.

All items are measured using a five-point Likert scale (1–5) referring to the previous short time window before the EMA prompt. Higher values indicate a stronger presence of the reported state.

The following variables were retained for analysis:

- *anxious* : momentary level of anxiety or emotional tension
- *calm* : perceived emotional calmness and stability
- *enthusiastic*: level of activation, energy, and enthusiasm
- *sympathetic*: degree of warmth and emotional support toward others
- *disorganized*: perceived lack of structure or care at that moment
- *reserved*: tendency toward quietness or low social expressiveness
- *dependable*: perceived self-discipline and reliability in recent behaviour

Responses were converted to numeric format and aggregated at the daily user level by computing the mean value of each variable when multiple responses were available within the same day. This aggregation produces stable daily indicators of momentary behavioural and emotional states while preserving within-person variability across days.

2.5.6 Comment

The Comment EMA module contains short free-text responses where participants can report what is currently on their mind. Unlike other EMA instruments, this block does not use ordinal or numeric scales, but open-ended textual input.

The core variable retained from this module is:

- *comment*: open-text response to the prompt asking participants what they are thinking about at the moment of the EMA entry.

Due to the unstructured and highly variable nature of these responses, no semantic or sentiment features were derived in this project. Instead, preprocessing focused on retaining only non-empty comment entries by removing missing values, trimming whitespace, and discarding blank responses. Empty or blank responses were therefore excluded from the analysis.

From this module, a single daily quantitative indicator was derived:

- *comment_count*: number of non-empty comment responses recorded for a given user on a given day.

This variable is used as a simple indicator of expressive engagement and EMA participation intensity, without analysing the linguistic content of the text itself, thereby avoiding additional complexity and privacy concerns.

2.5.7 Dimensions

The Dimensions EMA module consists of an open-ended descriptive question designed to capture participants' qualitative perceptions about their current context or experience. Responses are provided in free-text format and are not measured on a numeric scale.

The retained variable in this block is:

- *describe*: open-text response where participants describe their perceived situation or context in their own words.

Given its qualitative nature and variability in length and content, these responses were not transformed into semantic or psychological features. Instead, each valid response was treated as a single observation, and the total number of responses per user and day was used as a measure of interaction with the EMA system.

To obtain a consistent quantitative measure, the following daily variable was computed:

- *dimensions_count*: total number of valid Dimensions EMA responses recorded for each user on each day.

This variable serves as an indicator of daily EMA interaction level. Higher values reflect more frequent qualitative reporting, while zero values indicate no participation in this module on that day. It is therefore interpreted as a behavioural engagement signal rather than a direct psychological measure.

2.5.8 Dining Halls

The EMA Dining Halls module captures contextual information about where participants reported consuming their main meals. These variables provide structured location context that complements behavioural and routine indicators without directly measuring emotional states.

Participants reported the location of each main meal separately using three columns:

- breakfast
- lunch
- dinner

These fields were originally encoded using categorical numeric codes. During preprocessing, all codes were mapped to standardized human-readable categories defined in the dataset documentation, including campus dining locations such as Collis, Hop, Foco, KAF, Novack, East Wheelock Snack Bar, as well as Off-Campus and Other.

When multiple responses were available for the same user and day, the most frequently reported location was selected for each meal to obtain a stable daily representation.

After preprocessing and aggregation, the following daily variables were generated:

- *breakfast_place*: most frequently reported breakfast location
- *lunch_place*: most frequently reported lunch location
- *dinner_place*: most frequently reported dinner location

These variables provide categorical daily context about eating environments and are later used as complementary contextual features.

2.5.9 Event

The EMA Events module captures participants' self-reported evaluation of personally experienced events from the previous day. Specifically, participants were asked to recall their most positive and most negative events and rate their emotional intensity. Unlike momentary mood measures, this block focuses on important daily experiences instead of constant emotions. Two quantitative variables were retained:

- *positive*: reported intensity of the most positive event at the time of each EMA response.
- *negative*: reported intensity of the most negative event at the time of each EMA response.

Both variables are measured on a seven-point ordinal scale (1–7), where higher values indicate greater perceived emotional intensity.

Only responses that could be successfully converted to numeric values were retained. Non-numeric, missing, or invalid entries were discarded to ensure measurement consistency.

For daily modelling purposes, responses were summarized at the user-day level. When multiple entries were available on the same day, the following aggregated indicators were computed:

- *events_responses*: total number of valid Events EMA entries recorded that day.
- *positive_mean*: daily mean of positive event intensity scores.
- *negative_mean*: daily mean of the reported intensity for negative events.

Together, these variables provide a compact daily summary of emotional event intensity, complementing momentary EMA affect measures and passive sensing indicators.

2.5.10 Mood Variables

2.5.10.1 Momentary Mood

The Momentary Mood EMA module captures participants' momentary emotional state at the time of response. This instrument focuses on current affect, providing a near real-time snapshot of emotional well-being.

Two variables were retained:

- *happy*: self-reported level of happiness, measured on a four-point ordinal scale ranging from 1 ("a little bit") to 4 ("extremely").
- *sad*: self-reported level of sadness, measured on the same four-point ordinal scale, where higher values indicate greater sadness.

Only valid numeric responses were retained. For each user and day, multiple responses were summarized using the daily mean to obtain stable indicators of positive and negative affect.

The resulting daily variables are:

- *happy*: daily mean happiness level
- *sad*: daily mean sadness level

These variables provide core indicators of daily emotional state and serve as primary affective measures in subsequent analyses.

2.5.10.2 Expected Mood

The Expected Mood EMA module captures participants' short-term emotional expectations, introducing a prospective component to the assessment of well-being.

The retained variable is:

- *tomorrow*: expected emotional state for the following day, measured as a categorical variable with three possible responses: happy, stressed, or tired. Responses were encoded numerically (1–3) according to the questionnaire definition. When multiple responses were available for the same user and day, values were aggregated using the daily mean to obtain a single daily-level indicator. Although the resulting value does not correspond to a specific category, it provides a continuous approximation of the dominant expected emotional state (e.g., values closer to 1 indicate a tendency toward "happy", while values closer to 3 indicate a tendency toward "tired").

This feature complements current mood measures by incorporating perceived emotional outlook.

2.5.10.3 Global Mood

The Global Mood EMA module provides a compact global self-assessment of participants' overall emotional state, acting as a summary indicator of momentary well-being.

The retained variable is:

- *how*: current emotional state, with three possible responses: happy, stressed, or tired, encoded numerically (1–3) according to the questionnaire definition. When multiple responses were recorded within the same day, values were aggregated using the daily mean to obtain a single daily-level indicator. Although the resulting value does not correspond to a specific category, it provides a continuous approximation of the dominant emotional state at that moment (e.g., values closer to 1 indicate a tendency toward “happy”, while values closer to 3 indicate a tendency toward “tired”).

This variable serves as a concise global affect indicator that can be analysed alongside more specific mood and behavioural EMA measures.

2.5.11 Sleep

The Sleep EMA module captures self-reported information about sleep duration and perceived sleep quality for the previous night. These measures provide behavioural context relevant to recovery and daily functioning.

Two variables were retained:

- *hour*: number of hours slept during the previous night, reported as a continuous numeric value.
- *rate*: subjective sleep quality measured on a four-point ordinal scale ranging from 1 (very good) to 4 (very bad).

Only valid numeric responses were retained. When multiple records were available for the same user and day, values were aggregated using the daily mean.

The resulting daily variables are:

- *sleep_hours*: daily mean hours slept.
- *sleep_quality*: daily mean sleep quality score.

These variables provide stable daily indicators of sleep behaviour and perceived rest quality. In days without EMA responses, values were kept as missing (NaN), since these variables are self-reported and the absence of data cannot be interpreted as zero sleep or valid duration.

2.5.12 Social

The Social EMA module measures daily social exposure by recording how many people the participant interacted with during the day. This variable reflects behavioural social activity rather than perceived relationship quality.

The retained variable is:

- *social_people*: number of people interacted with, recorded as an ordinal categorical value encoded numerically in six levels (from low to high interaction range).

Invalid or missing responses were excluded. When multiple responses were available for the same user and day, the daily mean value was computed.

The resulting daily variable is:

- *social_people*: daily mean social interaction level.

This measure serves as an indicator of daily social engagement.

2.5.13 Stress

The Stress EMA module captures participants' perceived stress level, providing a subjective measure of psychological stress.

The retained variable is:

- *stress_level*: momentary self-reported emotional state measured on a five-point ordinal EMA scale ranging from 1 ("a little stressed") to 5 ("feeling great"), capturing a range from stress to positive well-being.

Only valid numeric responses were kept. When multiple entries were available for the same user and day, values were aggregated using the daily mean.

The resulting daily variable is:

- *stress_level*: daily mean perceived stress level.

This variable acts as a compact daily indicator of subjective stress and complements mood and behavioural EMA measures.

2.5.14 Study Spaces

The Study Spaces EMA module captures contextual information about participants' study environment and their perceived productivity while studying. This module provides complementary contextual indicators that can be analysed alongside mood, stress, and behavioural variables.

Two variables were retained from this module:

- *study_place*: categorical variable indicating the location where the participant was studying at the time of the EMA response (e.g., library, dorm, home, classroom, or other locations).
- *study_productivity*: self-reported perceived productivity while studying, measured on a four-point ordinal scale ranging from very unproductive to very productive.

Only valid responses were retained, and productivity scores were converted to numeric format for aggregation. Daily summaries were computed per user. When multiple responses were available on the same day, the daily mean productivity score was calculated. For study location, a representative daily value was selected based on the first valid reported location of the day to provide a stable contextual indicator.

The resulting daily variables are:

- *study_productivity*: daily mean perceived productivity while studying.
- *study_place*: representative study location for that day.

These variables provide contextual insight into daily study conditions and perceived performance, supporting the interpretation of emotional and behavioural patterns.

2.6 Passive Smartphone Sensing Processing

2.6.1 Sensing Data Role and Selection Criteria

Passive smartphone sensing data were incorporated as an objective behavioural layer that complements EMA self-reports and academic context variables. These data are generated automatically by device sensors and system logs, allowing continuous observation of daily routines and interaction patterns without requiring active user input.

Only sensing modalities that provide stable, recurring, and interpretable daily behavioural indicators were retained. The selection prioritized signals that can be reliably aggregated at daily level and that add complementary behavioural information to the analytical framework.

2.6.1.1 Excluded Sensing Modules

Only a subset of the available sensing modalities was used to ensure interpretability and alignment with the objectives of the project. Sensing modules such as GPS, WiFi location, and conversation were excluded due to their higher processing complexity, limited interpretability, or redundancy with other contextual signals already considered (such as physical activity, Bluetooth proximity, and device usage). Non-selected sensing data were therefore omitted to reduce noise and maintain a coherent and robust set of daily behavioural indicators.

2.6.2 Activity Inference

The Activity Inference sensing module captures participants' physical activity states automatically inferred from smartphone accelerometer signals. Each sensing record corresponds to a detected activity label at a given moment in time.

According to the dataset specification, activity inference is encoded as a categorical numeric variable with four possible values: stationary, walking, running, and unknown. Only valid activity labels were retained for analysis.

To obtain interpretable behavioural indicators, the original activity categories were recoded into two binary variables. The first indicator, *is_stationary*, takes value 1 when the inferred activity corresponds to a stationary state and 0 otherwise. The second indicator, *is_active*, takes value 1 when the inferred activity corresponds to walking or running, and 0 otherwise.

These binary indicators were then summarized at the daily user level by computing their mean value across sensing observations.

This produces two daily features: *prop_stationary* and *prop_active*, representing the proportion of the day classified as stationary and physically active, respectively.

Both variables take values between 0 and 1 and provide a robust daily-level summary of physical activity patterns while reducing short-term sensor noise and high-frequency fluctuations.

2.6.3 Audio

The audio sensing module provides contextual information about the participant's surrounding acoustic environment using automatic audio inference rather than raw recordings, thereby preserving privacy. Each sensing record is classified into general categories such as silence, human voice, noise, or unknown.

Only valid audio inference labels were retained. To obtain interpretable daily indicators, two binary variables were derived: *is_voice*, indicating detection of human voice, and *is_noise*, indicating detection of environmental noise.

These indicators were summarized at the daily user level by computing their mean across sensing observations.

This produced two proportion-based features: *prop_voice* and *prop_noise*, representing the fraction of the day in which voice or noise was detected.

Both variables are bounded between 0 and 1 and provide a lightweight daily summary of acoustic context and social-environmental exposure without relying on content-level audio analysis.

2.6.4 Bluetooth

The Bluetooth sensing module provides information about proximity to nearby devices and is used as an indicator of social density and co-presence. Bluetooth scans detect discoverable devices in the surrounding environment without identifying individuals, offering a privacy-preserving contextual signal. Only valid scan records were retained.

Two complementary daily indicators were derived from these data: *bt_unique_devices*, defined as the number of distinct devices detected during the day, and *bt_scans*, defined as the total number of Bluetooth scan events recorded.

These variables capture different aspects of proximity sensing. The number of unique devices reflects environmental or social diversity, while the number of scans provides a measure of sensing intensity and coverage for that day. Interpreting both measures jointly helps distinguish between low detected proximity due to true isolation and low values caused by limited sensing activity.

Together, these features provide a robust daily summary of proximity patterns and complement other behavioural and contextual sensing indicators.

2.6.5 Dark Screen

The dark screen sensing module captures time intervals during which the smartphone screen remained off, providing an objective indicator of device inactivity and reduced phone engagement. These intervals can be associated with sleep, rest periods, or sustained non-phone activities.

Only valid intervals were retained, and their durations were computed and aggregated at the daily user level by summing total screen-off time. The aggregated duration was converted to minutes to improve interpretability.

The resulting daily variable, *dark_minutes*, represents the total number of minutes per day during which the smartphone screen was off. This feature offers a general but informative indicator of daily disengagement from the device and complements activity and usage-based sensing measures.

2.6.6 Phone Charging

The phone charge sensing module records time intervals during which the smartphone was connected to a power source. This signal provides an indirect indicator of daily device usage patterns, routine structure, and charging habits.

Only valid charging intervals were retained, and their durations were computed and aggregated at the daily user level by summing total charging time across the day. The aggregated value was expressed in minutes to improve interpretability.

The resulting daily variable, *charge_minutes*, represents the total time per day during which the device was charging and serves as an indicator related to device engagement and routine regularity, complementing other phone usage sensing indicators.

2.6.7 Phone Lock State

The phone lock sensing module records time intervals during which the smartphone remained locked, providing an objective indicator of periods without direct device interaction. This signal complements other device usage measures by detecting periods when the smartphone was not being used, focusing on device lock status rather than only screen inactivity.

Only valid lock intervals were considered, and their durations were computed and aggregated at the daily user level by summing total locked time across the day. The aggregated duration was expressed in minutes for consistency with other device-state indicators.

The resulting daily variable, *locked_minutes*, represents the total number of minutes per day during which the device remained locked and serves as an indicator of reduced phone interaction, complementing screen state, charging, and activity sensing measures.

2.6.8 Wi-Fi

The Wi-Fi sensing module captures information about the surrounding environment through periodic scans of nearby access points. Only valid scan records were retained. To obtain a stable daily indicator, Wi-Fi data were summarized at the user level by counting the number of distinct access points detected per day.

The resulting variable, *unique_wifi*, represents the number of unique Wi-Fi access points observed during the day and serves as an indicator of environmental diversity and mobility. Higher values typically indicate exposure to multiple locations or network contexts, while lower values suggest more stable environments. This indicator complements other sensing measures related to activity, proximity, and routine patterns.

2.6.9 SMS Messages

The SMS sensing module captures participants' text messaging activity and was used as a passive indicator of digital social interaction. Only message event counts were considered, without analysing message content or communication partners.

Only valid SMS event records were retained. Daily indicators were obtained by counting the total number of SMS events recorded per user and day.

The resulting variable, *num_sms*, represents the daily number of text message events and provides a simple and interpretable measure of messaging activity. Although it does not reflect semantic or emotional content, it is a useful indicator of communication engagement and complements other sensing indicators related to proximity, mobility, and device usage.

2.7 Survey Data Processing

2.7.1 Survey Data Role and Temporal Treatment

Survey-based measures were incorporated as individual-level characteristics describing relatively stable psychological and personal attributes of participants. These measures are obtained through structured self-report questionnaires and represent baseline personal profiles rather than repeated daily observations.

Only pre-survey (“pre”) responses were retained when both pre- and post-assessments were available, ensuring that survey variables reflect baseline participant characteristics rather than later changes over time. Accordingly, survey variables were processed and retained at the user level and were not temporally aggregated. Each participant is therefore associated with a single value per survey-derived variable. The resulting survey features were integrated into the analytical dataset as time-invariant attributes, providing individual-difference context for subsequent analyses.

2.7.2 Excluded Questionnaires

The PSQI questionnaire was explored but excluded from the final analytical dataset due to inconsistencies in textual coding and lack of standardization.

The VR-12 questionnaire was also excluded because it mainly measures general health-related quality of life and physical functioning. Since this project focuses on daily emotional well-being, behavioural patterns, and psychological variables, the VR-12 was considered less directly relevant. In addition, the psychological and emotional dimensions central to this study were already covered by the other selected questionnaires.

2.7.3 Big Five Personality (BFI-44)

Personality traits were obtained using the Big Five Inventory (BFI-44) included in the survey module of the dataset. This instrument consists of 44 self-report items (e.g., “I see myself as someone who is talkative”, “tends to find fault with others”, or “does a thorough job”) designed to measure five personality dimensions: extraversion, agreeableness, conscientiousness, neuroticism, and openness. Each item is rated on a five-point Likert scale ranging from strong disagreement to strong agreement.

During preprocessing, responses were loaded from the corresponding survey file, and item columns were standardized to a consistent format. Textual Likert responses were converted into numeric values from 1 to 5 according to the official scale definition.

Several BFI-44 items are reverse-worded. These items were reverse-coded so that higher numeric values consistently represent higher levels of the corresponding personality trait, ensuring comparability across items within each dimension.

After these transformations, items were grouped following the standard BFI-44 structure. For each participant, one score per personality dimension was computed as the mean of the associated items,

ignoring missing responses when necessary. This produced five continuous personality variables for each participant: *extraversion*, *agreeableness*, *conscientiousness*, *neuroticism*, and *openness*.

2.7.4 Flourishing Scale

Psychological well-being was assessed using the Flourishing Scale included in the survey module. This instrument consists of eight Likert-type items (e.g., “I lead a purposeful and meaningful life”, “my social relationships are supportive and rewarding”, or “I am engaged and interested in my daily activities”) that measure positive psychological functioning, purpose, engagement, and relationship quality. Higher scores indicate higher levels of psychological well-being. Item responses were converted to numeric values, and invalid entries were treated as missing.

The overall flourishing score was computed by summing the valid responses across the eight items while ignoring missing values. This produced a single flourishing score, named *flourishing_score*, for each participant and used as a baseline feature.

2.7.5 UCLA Loneliness Scale

Perceived loneliness was measured using the UCLA Loneliness Scale included in the survey module. This questionnaire contains 20 items designed to assess subjective feelings of social isolation and lack of social connection (e.g., “I lack companionship”, “there is no one I can turn to”, or “I feel part of a group of friends”). Item responses were encoded using a four-point Likert scale reflecting frequency of experience and converted to numeric values according to the scale definition.

To ensure that higher values consistently represent higher perceived loneliness, positively worded items reflecting social connectedness were reverse-coded prior to score computation.

A total loneliness score, named *loneliness_score*, was then computed for each participant by summing the numeric values of the 20 items, resulting in a single baseline loneliness indicator per user.

2.7.6 Positive and Negative Affect Schedule (PANAS)

The Positive and Negative Affect Schedule (PANAS) was used to obtain baseline emotional characteristics of each participant. Each item is rated on a 5-point Likert scale, and responses were converted to numeric values from 1 to 5. The instrument includes two independent item groups: positive affect items, such as “interested” and “enthusiastic”, and negative affect items, such as “distressed” and “irritable”. The instrument includes two independent item groups measuring positive affect and negative affect.

In this dataset, the PANAS implementation contains 18 items (9 positive and 9 negative). For each participant, two separate scores were computed by summing the valid responses within each item group:

- *positive_affect_score*: sum of positive affect items.
- *negative_affect_score*: sum of negative affect items.

Missing values were ignored in the summation. The resulting scores provide two independent baseline indicators of participants' affective disposition.

2.7.7 Perceived Stress Scale (PSS-10)

The Perceived Stress Scale (PSS-10) was used to measure participants' baseline perceived stress levels. The questionnaire consists of 10 items rated on a 5-point Likert scale (e.g., "In the last month, how often have you been upset because of something that happened unexpectedly?", "In the last month, how often have you felt that you were unable to control the important things in your life?", or "In the last month, how often have you felt nervous and stressed?"). Responses were converted to numeric values from 0 to 4 following the official scoring scheme, where higher values indicate higher perceived stress.

Positively worded items were reverse-scored using the standard transformation ($4 - \text{response}$) to ensure consistent direction across all items.

A total perceived stress score, named *stress_score*, was computed for each participant by summing the valid item responses. Missing values were ignored in the summation. The resulting score provides a single baseline stress indicator per participant.

2.7.8 Patient Health Questionnaire (PHQ-9)

The Patient Health Questionnaire-9 (PHQ-9) is a standardized self-report instrument used to assess the presence and frequency of depressive symptoms over a recent reference period. It consists of 9 items (e.g., "Little interest or pleasure in doing things", "Feeling down, depressed, or hopeless", "Trouble falling or staying asleep, or sleeping too much", or "Feeling tired or having little energy"), each corresponding to a core symptom of depression commonly used in clinical and research settings.

Each item is rated on a 4-point Likert scale reflecting symptom frequency. Responses were converted to numeric values from 0 to 3 following the official scoring scheme, where higher values indicate greater symptom frequency. All items are formulated in the same direction, so no reverse scoring was required.

A total PHQ-9 score, named *phq_score*, was computed for each participant by summing the valid item responses, resulting in an overall severity indicator of depressive symptoms. Missing values were ignored in the summation. The resulting score provides a single participant-level psychological measure used in the analytical dataset.

2.8 Final Data Integration and Cleaning

In the final stage of the data preparation process, all previously processed and validated data sources were integrated into a single unified analytical table. The integration was designed to preserve all valid user-date observations obtained from at least one processed source, ensuring that no real records were artificially generated or removed during merging.

Different sources were combined according to their natural analysis level. Variables derived from daily records were merged using the *(user_id, date)* key, while participant-level attributes obtained from questionnaires were merged using *user_id* only and propagated across the corresponding daily rows. This preserves each variable linked to its correct day and allows different types of data to be combined in the same table.

Only features that had passed prior source-specific cleaning and validation steps were included. At this stage, no additional feature transformations were introduced beyond structural alignment and type consistency checks.

After integration, the resulting table was stored as a clean, analysis-ready dataset. This final dataset provides a coherent foundation for subsequent modelling and statistical analysis by combining behavioural indicators, contextual measures, and stable individual-level characteristics in a consistent framework.

2.9 Structure of the Analytical Dataset

The final analytical dataset consists of 77 columns, including 75 analytical variables and 2 identifier fields (*user_id* and *date*). These variables are grouped into conceptual blocks according to their origin and analytical role. For clarity, only the variables relevant to the main analysis are presented, while the full set of variables is detailed in Appendix A.

Table 2. Dataset Identifier Variables

Variable	Block	Source	Description
<i>user_id</i>	Identifiers	Integrated dataset	Unique participant identifier
<i>date</i>	Identifiers	Integrated dataset	Daily observation date

Table 3. Behavioural Measure Variables

Variable	Block	Source	Description
<i>total_apps_used</i>	Behavioural measures	<i>app_usage</i>	Number of distinct apps used per day
<i>tiempo_uso_total</i>	Behavioural measures	<i>app_usage</i>	Total app usage activity
<i>calendar_event_count</i>	Behavioural measures	<i>calendar</i>	Number of calendar events
<i>total_calls</i>	Behavioural measures	<i>call_log</i>	Total number of calls

Table 4. Academic and Educational Variables

Variable	Block	Source	Description
num_deadlines_today	Academic measures	deadlines.csv	Number of deadlines

Table 5. EMA Emotional Variables

Variable	Block	Source	Description
happy	EMA emotional	Mood	Daily happiness
stress_level	EMA emotional	Stress	Daily stress level

Table 6. EMA Contextual Variables

Variable	Block	Source	Description
hours	EMA contextual	Class	Study hours
sleep_hours	EMA contextual	Sleep	Sleep duration
sleep_quality	EMA contextual	Sleep	Sleep quality
social_people	EMA contextual	Social	Social exposure

Table 7. Passive Sensing Variables

Variable	Block	Source	Description
prop_stationary	Passive sensing	activity inference	Time stationary
prop_active	Passive sensing	activity inference	Time active
bt_unique_devices	Passive sensing	bluetooth	Unique devices
dark_minutes	Passive sensing	dark	Screen inactivity
locked_minutes	Passive sensing	phonelock	Locked time
unique_wifi	Passive sensing	wifi	WiFi diversity
num_sms	Passive sensing	sms	SMS activity

Table 8. Survey-Based Psychological Variables

Variable	Block	Source	Description
extraversion	Survey traits	BigFive	Extraversion
agreeableness	Survey traits	BigFive	Agreeableness
conscientiousness	Survey traits	BigFive	Conscientiousness
neuroticism	Survey traits	BigFive	Neuroticism
openness	Survey traits	BigFive	Openness
flourishing_score	Survey traits	Flourishing	Well-being
loneliness_score	Survey traits	Loneliness	Loneliness
panas_positive	Survey traits	PANAS	Positive affect
panas_negative	Survey traits	PANAS	Negative affect
stress_score	Survey traits	PSS	Perceived stress
phq_score	Survey traits	PHQ-9	Depression score

2.10 Chapter Summary

This chapter described the data preparation process applied to the StudentLife dataset. After introducing the dataset and its main sources, a unified preprocessing strategy was defined, including timestamp normalization to the local time zone, aggregation at the user–day level, and the handling of missing and inconsistent records.

Each data source was then processed according to its nature: behavioural logs, academic information, EMA responses, passive smartphone sensing data, and survey-based questionnaires. All processed sources were integrated into a single analytical dataset combining daily-level and participant-level variables, resulting in a clean and structured table ready for the subsequent exploratory and analytical stages of the project.

3 EXPLORATORY EMOTIONAL ANALYSIS

Figure 7 provides an overview of the analytical structure of this chapter.

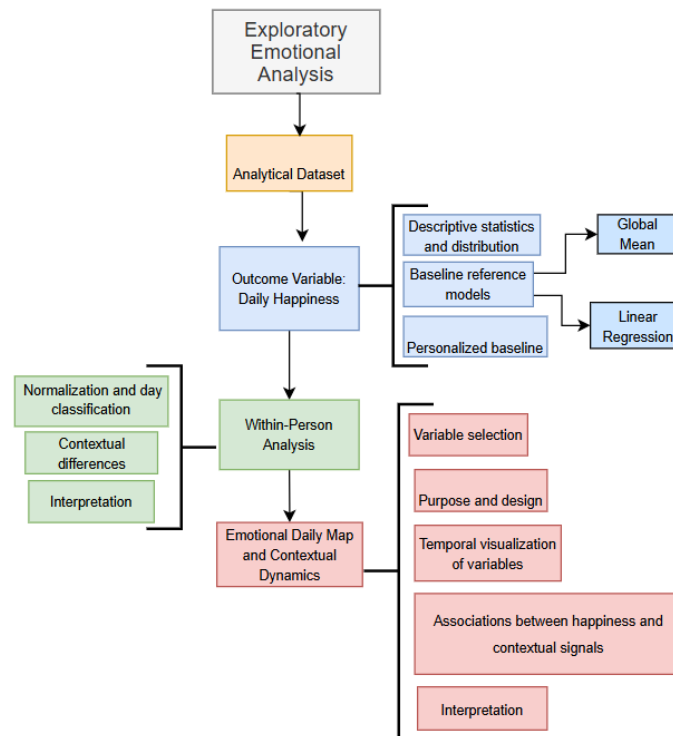


Figure 7. Analytical Workflow of the Exploratory Emotional Analysis

3.1 Dataset for Emotional Analysis

Following the data preparation and integration process described in Chapter 2, the resulting analytical dataset was used as the basis for the exploratory emotional analysis presented in this chapter. The final dataset consists of 4,753 daily user-level observations corresponding to 49 participants and 184 distinct days of monitoring. This number differs slightly from the 48 participants reported in the original StudentLife study [4], likely reflecting differences between the publicly available dataset and the subset of participants used in the analyses reported in the paper. Figure 8 summarizes the number of participants.

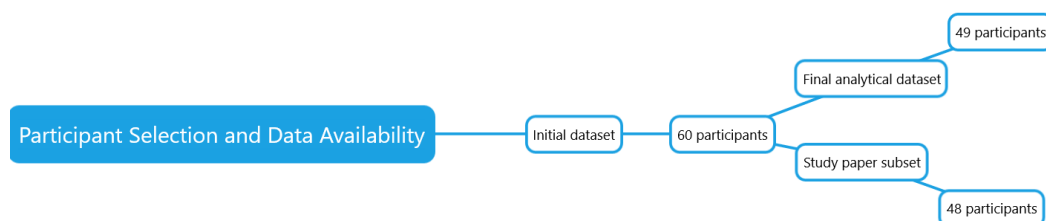


Figure 8. Participant Selection and Data Availability

The final analytical dataset consisted of 77 columns, including 75 analytical variables and 2 identifier fields (`user_id` and `date`). These variables combined passive sensing data, activity logs, EMA questionnaires, and psychological scales. No duplicated rows were detected after daily aggregation. Missing values are present across variables, but are more prevalent in EMA-related measures, since these data were only recorded when participants actively responded and are therefore not available for all user–day observations.

For each specific analysis in this chapter, smaller working datasets were derived from the final analytical dataset by selecting only the variables required for that particular step. This was done to keep each analysis focused and to avoid unnecessarily discarding observations due to missing values in variables that were not relevant to that analysis. As a result, the number of observations may vary across different sections of this chapter.

During the data exploration stage, user identifiers were found to range up to 60, suggesting that the original dataset was designed to include up to 60 participants. However, not all identifiers were present in the processed dataset. Several intermediate IDs were missing because no valid observations were available for these users in the raw data. Specifically, identifiers 6, 11, 21, 26, 28, 29, 37, 38, 40, 48, and 55 did not appear after preprocessing and aggregation, and therefore they could not be included in the final analysis. As a result, the final analysis was conducted using 49 participants.

The analytical dataset described above provides the basis for the exploratory emotional analysis conducted in this chapter. The following sections describe the selected outcome variable and its main properties. The analysis begins by defining this variable and assessing its statistical properties, before introducing simple baseline models and progressively more structured comparisons.

3.2 Daily Happiness as Outcome Variable

The target variable selected for the exploratory emotional analysis is the daily happiness score obtained from EMA mood reports. This variable was chosen because it directly reflects participants' self-reported emotional state in everyday situations and provides a concrete indicator of subjective well-being at the daily level.

Unlike baseline psychological scales, which represent relatively stable personal traits, the happiness EMA measure captures short-term emotional variation. This makes it especially suitable for analysing within-person emotional dynamics and relating daily behavioural and contextual factors to changes in reported well-being. Happiness responses are recorded on an ordinal 4-point intensity scale ranging from 1 (low happiness) to 4 (high happiness). Higher values indicate stronger momentary positive affect. This variable is therefore used as the main dependent outcome in the analyses presented in this chapter.

While daily happiness serves as the primary outcome variable, the analysis does not rely on this measure in isolation but considers additional variables to provide a broader understanding of daily emotional dynamics.

3.2.1 Descriptive Statistics and Distribution

To better understand the availability and statistical properties of this outcome variable, an initial descriptive analysis was conducted. After daily aggregation, a total of 220 user-day observations with valid happiness reports were available (i.e., data are available). These observations correspond to 38 participants and 47 distinct calendar days, confirming the irregular and sparse nature of EMA mood responses across users and time. This descriptive step is necessary to assess the reliability of the happiness variable and to account for the sparse and irregular nature of EMA data, ensuring that subsequent analyses are interpreted within the appropriate data context.

The average happiness score across valid observations was 1.79 (SD = 0.83), with a median value of 2 on the 1–4 scale, indicating noticeable variability in reported happiness relative to the limited range of the scale. These statistics were computed using the subset of user-day observations with valid happiness values after filtering out missing EMA responses. The distribution is skewed toward lower values, with most reports concentrated at levels 1 and 2, while higher values (3 and 4) appear less frequently. This indicates that high momentary happiness states are comparatively rare in the recorded EMA responses.

3.2.2 Baseline Reference Models

To provide a reference point for subsequent analyses, several simple baseline prediction strategies were evaluated. These baseline models serve as lower-complexity reference points against which more context-based approaches can be compared.

First, a global mean baseline was defined by predicting the same value for all observations, equal to the overall mean happiness score. In this case, the constant prediction value was 1.79. Using this approach resulted in a mean absolute error (MAE) of 0.71. MAE measures the average absolute difference between the observed happiness values and the predicted values. In this case, it represents the average deviation between real happiness scores and a prediction that ignores both user identity and daily context.

Second, to test whether using contextual and behavioural variables improves prediction compared to a simple baseline, a linear regression model was trained using sleep-related measures, a phone usage indicator reflecting overall daily device activity, perceived stress, and social interaction variables as predictors. These features were selected because sleep, stress, social exposure, and device usage are commonly linked to short-term emotional variation in daily-life studies. Only days with complete data across the selected variables were retained, resulting in 65 observations from 13 participants. The regression model resulted in a prediction error similar to the global mean baseline. This suggests that although daily behavioural and contextual signals are related to emotional variation, their predictive power remains limited in this dataset, possibly due to the small number of complete observations and the irregular nature of EMA responses.

3.2.3 Personalized Baseline Comparison

To account for stable individual differences in reported happiness levels, a participant-specific baseline was computed by predicting each observation using that participant's own mean happiness across their available days.

This simple personalization strategy reduced the prediction error to an MAE of 0.53, substantially improving upon both the global mean baseline and the contextual linear regression model. The comparison across baseline approaches indicates that between-person differences explain a large proportion of the variance in happiness scores.

Consequently, the subsequent analyses focus primarily on within-person comparisons, examining how daily contextual factors relate to deviations from each participant's typical happiness level rather than relying only on cross-participant differences. These findings should be interpreted with caution due to the limited number of complete EMA observations and the inherent sparsity of daily self-reports.

To provide a clearer understanding of how these baseline approaches differ, the following simplified example (Figure 9) illustrates their behaviour across three users.

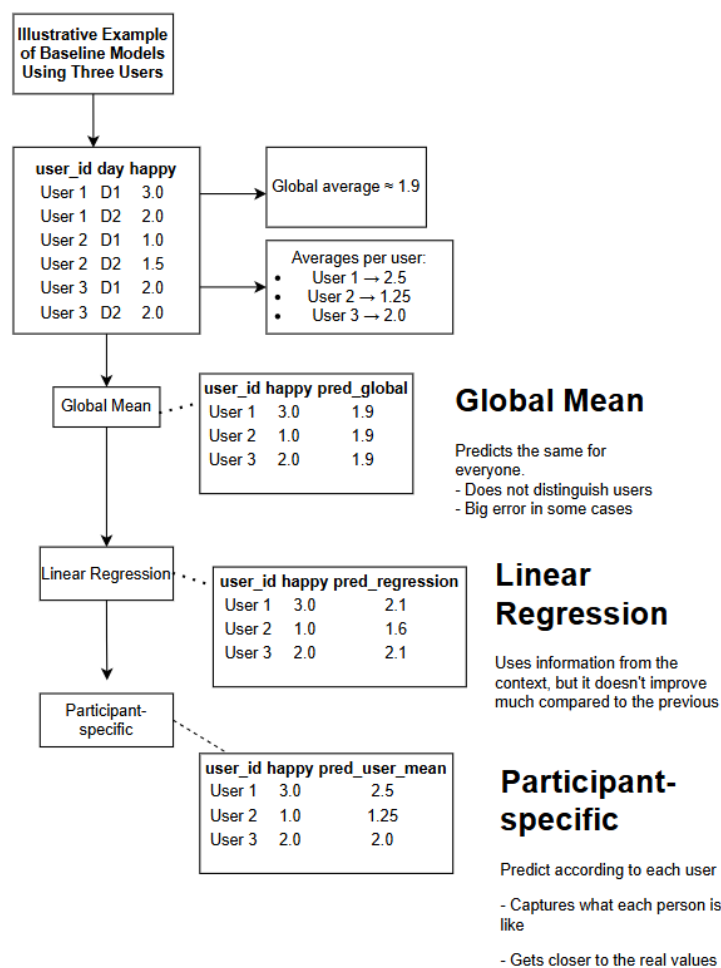


Figure 9. Illustrative Comparison of Baseline Models Across Three Users

3.3 Within-Person Analysis of Daily Happiness

The objective of this analysis is to examine whether daily contextual and behavioural patterns differ between days when a participant reports higher versus lower happiness relative to their own average. A within-person approach was selected because previous results showed that stable individual differences explain a large proportion of happiness variance. Therefore, daily scores are interpreted relative to each participant's own baseline instead of being compared directly across users. This allows contextual effects to be studied at the day level while controlling for individual reporting tendencies.

3.3.1 Within-person Normalization and Day Classification

To account for individual differences in baseline happiness, each day was evaluated relative to the participant's own average happiness across available days. This step was included because participants differ in their typical reporting levels, and direct cross-person comparisons would mix individual differences with day-level variation. By normalizing relative to each participant's own mean, the analysis focuses on deviations from personal baseline rather than absolute scores.

After removing days where all contextual variables were missing, this analysis was conducted on 220 user-day observations from 38 participants. The same variables used in the linear regression baseline were retained for this within-person analysis: sleep duration, sleep quality, number of apps used, total smartphone usage time, social interaction, and perceived stress.

Days were then classified as better-than-usual or at-or-below-usual by comparing the reported happiness score (happy) with the individual's personal mean (happy user mean). Across all observations, 72 days were classified as better-than-usual and 148 days as at-or-below-usual, indicating that days with higher-than-typical happiness occurred less frequently than neutral or lower days. This binary labelling strategy was chosen because it produces two interpretable and statistically stable groups that allow contextual comparison while remaining robust given the limited EMA sample size.

To better illustrate how this classification works, a simplified example is presented below (Table 9).

Table 9. Illustration of better-than-usual Day Classification

user id	day	happy	happy user mean	better day
User 1	D1	3.0	2.5	1
User 1	D2	2.0	2.5	0
User 1	D3	2.5	2.5	0
User 2	D1	1.5	1.8	0
User 2	D2	2.0	1.8	1

3.3.2 Contextual Differences Between Daily States

After classifying days relative to each participant's own happiness baseline, daily states were defined as better-than-usual and at-or-below-usual days. Daily contextual variables were then aggregated by computing the average value of each variable within these two groups. On better-than-usual days, participants showed higher average sleep duration (8.30 vs. 7.81 hours), greater mobile phone engagement both in overall device activity, measured as the total number of recorded usage events per day (1313 vs. 919) and number of apps used (16.15 vs. 13.34), as well as slightly higher perceived stress levels (2.80 vs. 2.36) and social interaction (3.63 vs. 3.34). Sleep quality, however, was marginally lower on better-than-usual days (1.65 vs. 1.92). These results indicate that higher-than-typical happiness days are associated with systematically different daily behavioural and contextual profiles rather than random fluctuation.

3.3.3 Magnitude and Interpretation of Contextual Differences

Examining the magnitude of the differences between day types further highlights the contextual patterns linked to daily well-being. Compared to at-or-below-usual days, better-than-usual days were characterized by substantially higher mobile phone activity, reflected in an increase of approximately 395 units in overall device activity and about 2.8 additional apps used, together with moderately longer sleep duration of around half an hour. Perceived stress and social interaction were also slightly higher, while sleep quality showed a small decrease. Taken together, these patterns suggest that days with higher-than-typical happiness are associated with higher levels of measured daily activity and device usage indicators, rather than simply with lower demands or greater rest. While these results should be interpreted with caution due to the limited and irregular EMA coverage, they are consistent with the observed contextual differences and support the use of within-person baseline comparisons when analysing short-term emotional variation.

3.4 Emotional Daily Map and Contextual Dynamics

3.4.1 Methodological Justification of Variable Selection

The Emotional Daily Map was constructed using a reduced set of daily variables selected to represent complementary dimensions of day-to-day context and emotional functioning. The selection was guided by interpretability and theoretical relevance for well-being analysis.

Daily happiness was used as the central emotional indicator because it provides a direct self-reported measure of momentary well-being and captures short-term emotional variation. Perceived stress was included as a key psychological context variable due to its well-established relationship with affective states.

Sleep duration and self-reported sleep quality were included as recovery-related contextual indicators because sleep is commonly associated with emotional regulation and daily functioning. However, in this sample, their relationship with daily happiness was not consistent, so they were retained mainly to represent an important lifestyle dimension of well-being.

The daily activity proportion derived from passive sensing was included as an objective behavioural activation measure. It reflects how active each day was and complements self-reported variables with sensor-based behavioural information. Additional indicators of smartphone engagement, such as overall device activity and the number of applications used per day, were also included to capture patterns of daily digital activity that may reflect behavioural engagement and routine.

Together, these variables provide a compact and interpretable representation of emotional state, perceived psychological load, recovery behaviour, digital engagement, and daily activity, supporting a within-person contextual analysis of emotional dynamics.

3.4.2 Purpose and Design of the Emotional Daily Map

Emotional well-being varies from day to day and is strongly influenced by daily context, making isolated emotional measurements insufficient. The Emotional Daily Map is introduced to capture temporal dynamics of emotional states and their relationship with behavioural and contextual factors such as sleep, stress, activity, and social interaction.

To illustrate this representation, an example participant (User 59) was selected, as this participant had the highest number of days with complete observations across the selected variables, allowing for a more reliable and interpretable visualization of daily patterns. This user provided 43 days with valid happiness EMA reports, and all of them contained contextual information after removing days where all contextual variables were missing. Based on the participant's personal mean happiness score (2.06), days were classified into 15 better-than-usual days and 28 at-or-below-usual days.

To further examine contextual differences at the individual level, average values of selected variables were compared between better-than-usual and worse-or-equal days for the same user. The results show that better days are associated with higher stress levels (+1.09), slightly longer sleep duration (+0.17 hours), increased smartphone usage both in total time (+115.46 minutes) and number of apps used (+0.44), and a higher proportion of active time (+0.009). In contrast, sleep quality appears slightly lower on better days (−0.27).

Overall, these differences suggest that improved emotional states are not necessarily linked to more relaxed or optimal conditions, but rather to specific combinations of contextual factors that vary within the individual.

This representation supports within-person interpretation by linking the results with visible day-to-day patterns. Case-based visualizations are commonly used in exploratory analyses of behavioural sensing data to illustrate how contextual signals and emotional states interact over time at the individual level. It complements the previous baseline and modelling results by adding a clear visual view that helps explain how contextual variables change together with emotional reports over time, instead of relying only on aggregated values.

3.4.3 Temporal Visualization of Emotional and Contextual Variables

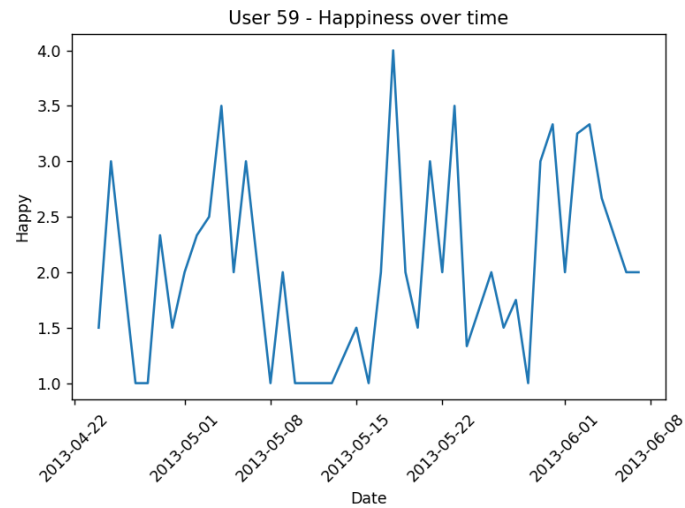


Figure 10. Daily Happiness Scores Over Time for User 59

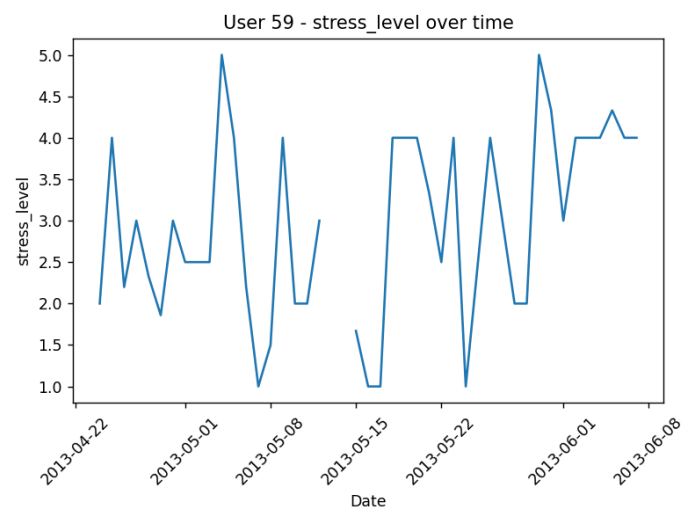


Figure 11. Daily Perceived Stress Level Over Time for User 59

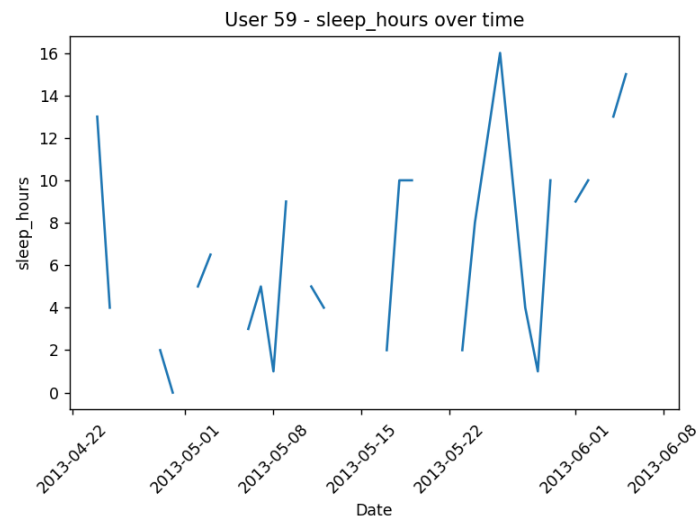


Figure 12. Daily Sleep Duration Over Time for User 59

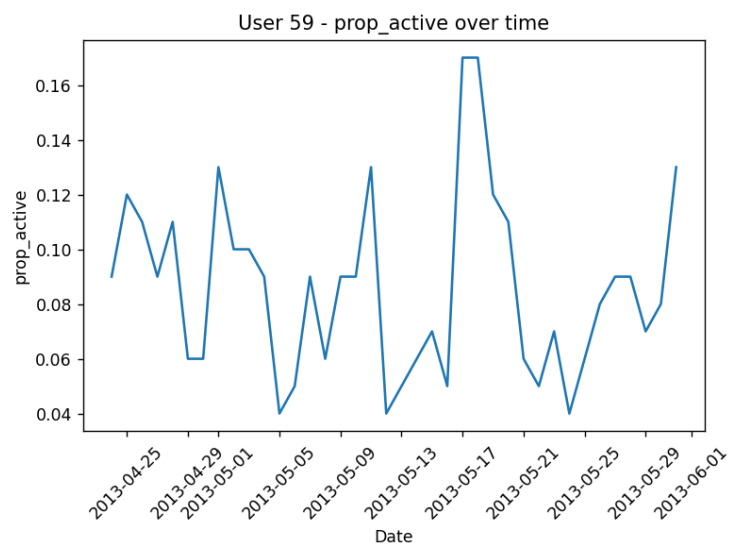


Figure 13. Proportion of Time Spent in Active States Over Time for User 59

Figure 10 shows the evolution of daily happiness scores over the observation period. Happiness exhibits substantial day-to-day variability rather than remaining stable, reinforcing the importance of analysing emotional well-being relative to each individual's own baseline rather than using absolute levels. Periods of higher and lower emotional states alternate across time, motivating the distinction between better-than-usual and worse-or-equal-than-usual days.

Figure 11 depicts perceived stress level over time. Stress fluctuates considerably and does not display a simple inverse relationship with happiness. Visually, several periods of higher happiness coincide with elevated stress levels. This observation is consistent with the comparison between better and worse days, where average stress level is notably higher on better-than-usual days (+1.09).

Figure 12 shows daily sleep duration. Sleep hours vary notably across days, with both short and unusually long nights. No clear temporal pattern is observed. Some apparent jumps in the curve are

partly due to missing daily records, since sleep data are not available every day and the plot connects the nearest available observations. Sleep is therefore retained as a contextual variable to preserve behavioural interpretability, rather than because it shows a strong isolated effect.

Figure 13 presents the proportion of time spent in active states, derived from passive sensing data. This variable shows noticeable day-to-day fluctuations, with no clear temporal pattern. Some variability reflects genuine changes in daily activity, while other irregularities may be influenced by gaps in the recorded data. Overall, the pattern suggests that activity levels vary considerably over time for this user.

The happiness plot includes all days with valid happiness responses. However, some contextual variables are not available for every one of those days. Therefore, their plots may contain fewer points or appear discontinuous due to missing data, even though they refer to the same observation period.

3.4.4 Within-Person Associations Between Happiness and Contextual Factors

Each point in the scatter plots represents a single day for the selected participant, relating the reported happiness level with the corresponding contextual variable measured on that day.

Happy vs Stress Level

Figure 14 suggests a slight positive tendency between perceived stress and daily happiness, although the relationship is not strong and shows considerable variability. Higher happiness values appear somewhat more frequently at moderate to high stress levels, but the pattern is not consistent across all observations. This observation is consistent with the within-person baseline comparison, where better-than-usual days for this user are associated with higher perceived stress levels. Overall, this indicates that higher engagement or demand days are not necessarily emotionally worse.

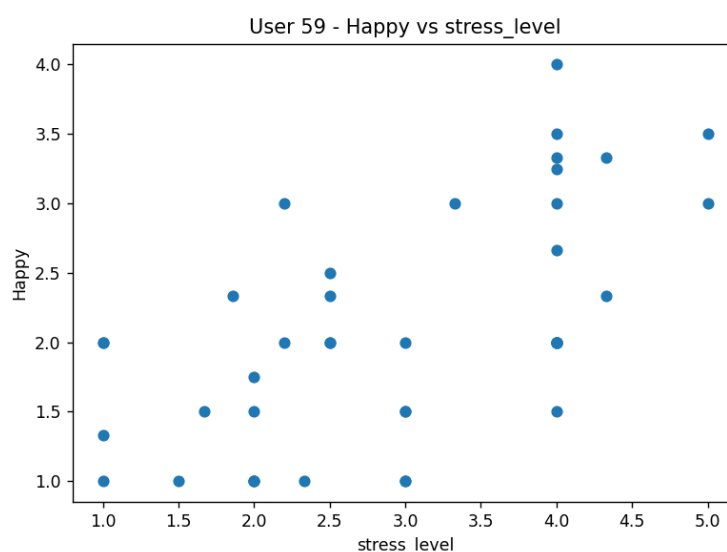


Figure 14. Relationship Between Daily Happiness and Perceived Stress Level for User 59

Happy vs prop_active

Figure 15 suggests a mild positive association between daily activity proportion and happiness for this user. Lower values of prop_active are more frequently associated with lower happiness scores, whereas higher activity proportions tend to coincide with higher happiness levels. Although the relationship is not strictly linear, the pattern indicates that more active days are generally associated with better emotional states for this individual.

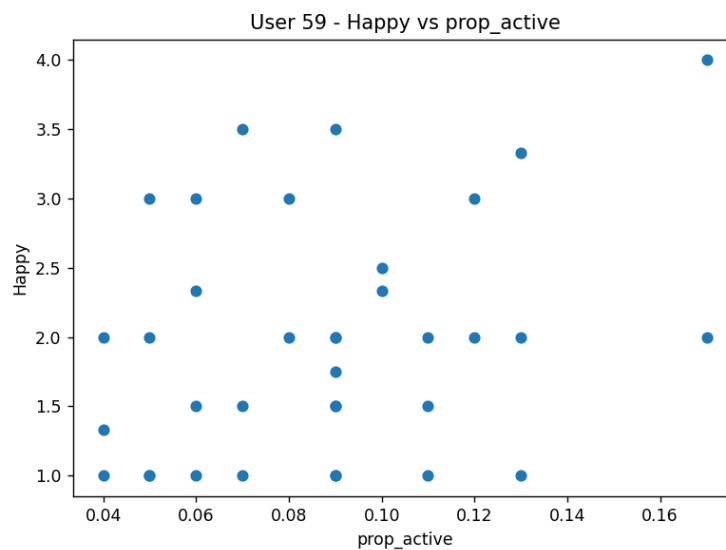


Figure 15. Relationship Between Daily Happiness and Proportion of Active Time for User 59

3.4.5 Interpretation

Using EMA and passive sensing data, a daily representation of each user's emotional state and contextual conditions has been constructed. The exploratory analysis reveals substantial intraindividual variability and person-specific contextual relationships, supporting the decision to adopt a within-person analytical framework and personalized baselines. These results reinforce the methodological choice of modelling well-being relative to each individual's typical level and motivate the use of personalized approaches in subsequent well-being modelling.

Since this visualization is based on a single illustrative participant, the patterns observed should be interpreted as exploratory examples of within-person dynamics rather than generalizable relationships across users.

3.5 Chapter Summary

This chapter presented the exploratory emotional analysis carried out on the integrated dataset. Daily happiness, obtained from EMA mood reports, was selected as the main outcome variable, and its statistical properties and distribution were described. Several baseline prediction strategies were

evaluated, including a global mean baseline, a linear regression model, and a personalized baseline based on each participant's own mean happiness.

A within-person analysis was then conducted by comparing each day's happiness score with the participant's personal baseline and classifying days as better-than-usual or at-or-below-usual. Contextual differences between these two day types were compared and interpreted. Finally, an Emotional Daily Map was constructed for an illustrative participant to visualize and explore daily emotional and contextual variables over time.

4 INDIVIDUAL EMOTIONAL VARIABILITY AND STABILITY

4.1 Conceptual Motivation and Analytical Rationale

In the previous chapter, the exploratory emotional analysis suggested that daily happiness levels vary over time and across observations. These results suggest that emotional well-being cannot be fully understood by looking only at isolated observations or overall averages, but requires examining how emotional states evolve over time within each person.

Most emotional well-being analyses focus on average affect levels, but the mean alone does not capture how stable or variable emotional states are across days. Two individuals can show similar average happiness while differing greatly in the stability of their daily emotional experiences. For this reason, this chapter separates emotional level (mean happiness) from emotional stability (day-to-day variability).

Emotional stability is analysed because the project follows a personalized well-being perspective, where not only the typical emotional level but also its predictability is relevant. Variability is therefore treated as a complementary dimension that adds interpretative value beyond averages. Since variability is inherently a within-person property, the analysis is conducted at the individual level. The objective is to explore individual emotional variability alongside average happiness levels to obtain a more complete descriptive view of daily well-being patterns.

4.2 Outcome Definition: Daily Happiness

Daily EMA happiness is used as the emotional outcome measure. It is retained here because it provides a day-level self-reported affect indicator suitable for within-person variability analysis. Its construction and preprocessing were described in previous chapters.

4.3 Inclusion Criteria and Data Reliability Constraints

To ensure that emotional variability can be estimated with sufficient reliability, a minimum availability condition was applied to daily happiness records. A valid observation was defined as a day with a non-missing happiness (EMA) report.

While general data cleaning and preprocessing were performed during dataset construction, additional filtering steps were applied at this stage to construct analysis-specific subsets.

Since variability is assessed within each user by examining day-to-day changes in happiness, only users with a valid threshold were retained for the analysis. This resulted in a filtered subset of the dataset specifically designed for estimating within-person emotional variability.

4.3.1 Minimum Observation Threshold

Users were required to have at least three valid daily EMA happiness reports. This requirement was set because variability metrics cannot be meaningfully estimated from very small numbers of observations.

4.3.2 Final Analytical Sample

After applying this criterion, 21 users were retained for the variability analysis. The study remains descriptive and exploratory in scope.

4.4 Stability Metrics Definition and Justification

Two per-user summary metrics were used: mean daily happiness and the within-user standard deviation of happiness. The mean represents overall emotional level, while the standard deviation captures day-to-day variability.

Standard deviation was selected because it provides a clear and interpretable measure of within-person fluctuation and allows emotional stability to be compared consistently across users.

4.5 Cross-User Variability Analysis

To visualize how emotional level and variability differ across individuals, a scatter plot of mean happiness and within-user standard deviation was constructed.

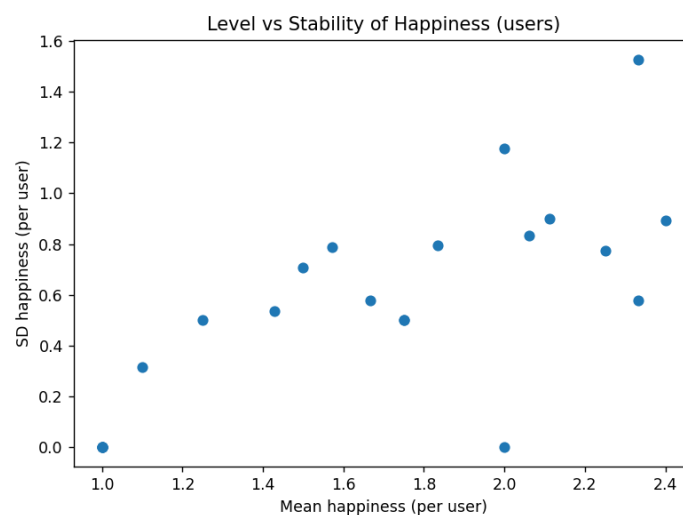


Figure 16. Mean Happiness vs Emotional Variability Across Users

Figure 16 shows the relationship between mean happiness and emotional variability across users using per-user averages and standard deviations. The scatter plot highlights substantial differences in stability: users with similar mean happiness can exhibit very different levels of variability. Although 21

users are included in the analysis, some points overlap in the plot because multiple users share identical or very similar mean and standard deviation values, resulting in only 17 visually distinguishable points.

Some users exhibit almost no day-to-day variation, while others show pronounced fluctuations. Higher average happiness does not necessarily correspond to greater stability. These results indicate that emotional level and emotional stability capture different aspects of well-being.

4.6 Identification of High-Variability Emotional Profiles

Building on the cross-user variability analysis, the following section identifies the individuals with the highest emotional variability (based on happiness SD).

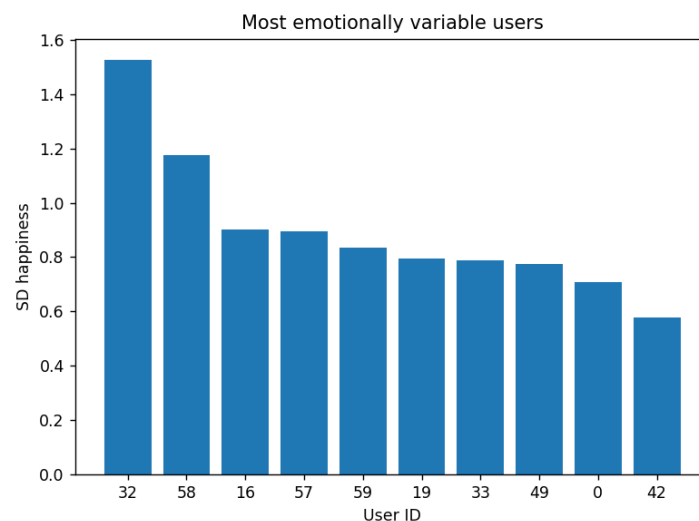


Figure 17. Users with Highest Emotional Variability Ranked by Happiness SD

Figure 17 presents the users with the highest emotional variability, ranked according to their standard deviation of daily happiness. The distribution shows that variability is not uniform across users but concentrated in a smaller subset with clearly higher dispersion values.

This ranking highlights that a limited group of user's accounts for the strongest day-to-day emotional fluctuations observed in the dataset.

4.7 Chapter Summary

The results show that emotional level and emotional stability represent distinct aspects of well-being. Users with similar average happiness can display very different variability patterns, which indicates that mean values alone do not fully describe individual emotional dynamics. Considering both level and variability provides a more complete descriptive view of user emotional patterns within the scope of this exploratory analysis.

5 SURVEY-BASED PERSONALITY AND WELL-BEING PROFILES

Figure 18 provides an overview of the analytical structure and main steps involved in the construction and evaluation of survey-based personality and well-being profiles.

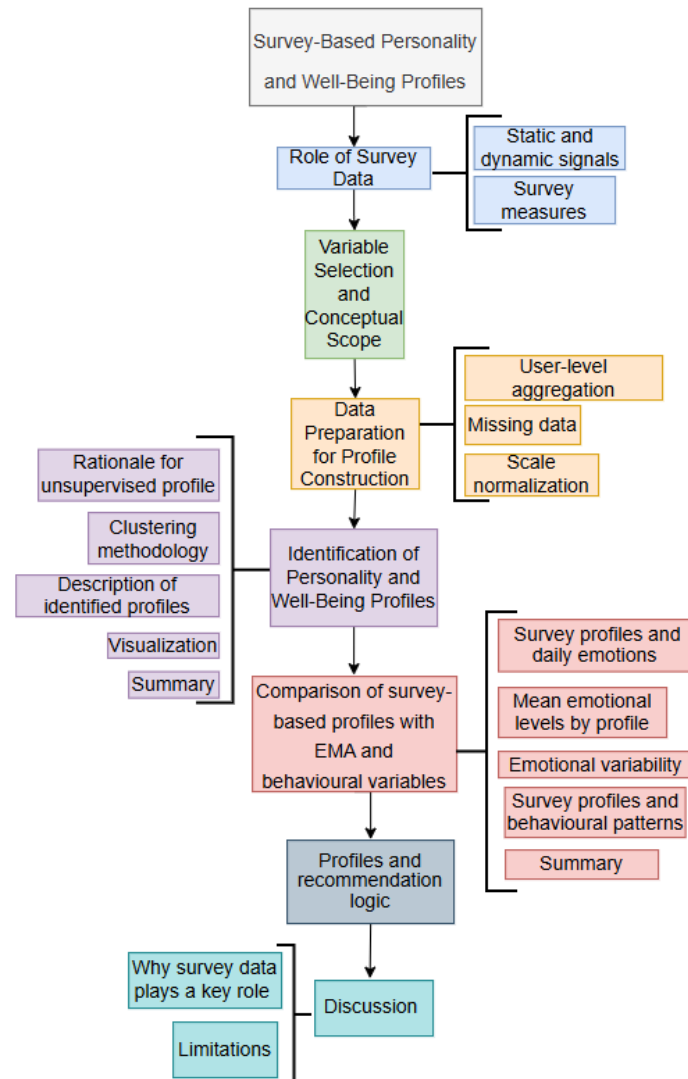


Figure 18. Analytical Workflow for Survey-Based Personality and Well-Being Profile Construction and Evaluation

5.1 Role of Survey Data as a Baseline for Well-Being Analysis

5.1.1 Static and Dynamic Signals in Well-Being Data

Well-being analysis typically relies on two complementary types of data: dynamic signals and static descriptors. Dynamic signals include daily self-reports and passively sensed behavioural data, which capture short-term emotional and contextual fluctuations. Although these signals provide detailed

temporal information, they are often affected by missing values, irregular sampling, and noise, which can complicate their interpretation.

Static descriptors, in contrast, represent relatively stable individual characteristics, such as personality traits and baseline well-being measures obtained through surveys. These variables change slowly over time and reflect enduring psychological tendencies rather than momentary states.

Distinguishing between dynamic and static signals is methodologically important, as dynamic data describe variability, while static measures provide a stable reference for interpreting individual differences in well-being patterns.

5.1.2 Survey Measures as a Stable Baseline for Personality and Well-Being

In this work, survey measures are used to define a stable personal profile for each user, based on personality traits and general well-being indicators. This profile summarizes long-term psychological characteristics that describe how each individual usually tends to feel and function.

This profile is later used to group users into interpretable profiles and to compare emotional and behavioural patterns across these groups. In this way, survey measures provide an individual baseline that supports subsequent profile-based and comparative analyses.

5.2 Survey Variable Selection and Conceptual Scope

5.2.1 Included Psychological Instruments

Survey variables were selected from the consolidated dataset focusing only on standardized self-report psychological measures. The selection focuses on survey-based variables that capture both stable personality traits and baseline psychological well-being. Specifically, it covers the Big Five personality dimensions: extraversion, agreeableness, conscientiousness, neuroticism, and openness, which describe individual behavioural tendencies. In addition, it includes psychological well-being measures such as flourishing (overall psychological well-being and sense of purpose), loneliness (perceived social isolation), affective balance from PANAS (tendency toward positive and negative emotions), perceived stress, and depressive symptoms measured through PHQ. Only these instruments were included to ensure conceptual coherence and to avoid mixing stable psychological measures with contextual or behavioural variables in the profile construction stage.

5.3 Survey Data Preparation for Profile Construction

5.3.1 User-Level Aggregation Strategy

The original dataset contains multiple daily records per user, while survey-based psychological measures remain constant for each individual. Therefore, survey variables were aggregated at the user level before profile construction.

This step ensures a single representation per user. The result is one survey-based feature vector per user used for profile identification.

5.3.2 Missing Data Handling

Before clustering, missing values in the survey variables were handled to ensure data quality and compatibility with the clustering algorithm.

While general data cleaning and preprocessing were performed during dataset construction, additional filtering steps were applied here to construct a reliable user-level dataset for profile identification.

Users with a high proportion of missing values were removed by applying a minimum availability threshold, excluding those with more than 20% missing responses across survey variables. This step ensures that only users with sufficiently complete survey information are included in the analysis.

For the remaining data, missing values were imputed using the median of each variable. This approach provides a neutral estimate that preserves the overall distribution of the data while avoiding distortions caused by extreme values.

5.3.3 Scale Normalization and Standardization

Before clustering, all survey variables were standardized using the z-score. This means that each variable was transformed separately across all participants by subtracting its mean and dividing by its standard deviation. As a result, each participant's value indicates whether they are above or below the average for that specific variable. This process puts all variables on the same scale, preventing variables with larger numeric ranges from dominating the clustering results and ensuring that all features contribute more equally.

5.4 Identification of Personality and Well-Being Profiles

5.4.1 Rationale for Unsupervised Profile Discovery

An unsupervised approach was used because the survey data did not include predefined profile labels. The objective was to identify natural groupings of users based on their personality and well-being measures.

Unsupervised clustering enables the discovery of data-driven profiles without imposing prior categories, supporting an exploratory and interpretable profile construction process.

5.4.2 Clustering Methodology

To identify personality and well-being profiles, a KMeans clustering algorithm was applied to the standardized survey variables. This method was selected because it provides clear and interpretable group structures based on similarity in multivariate feature space.

Different numbers of clusters were evaluated and compared using the silhouette coefficient. The configuration with two clusters produced the highest silhouette score (*0.19*), compared to *0.15* for $k = 3$ and *0.12* for $k = 4$, and was therefore selected as the final profile solution. Although the absolute value is relatively low, this is expected given the continuous and overlapping nature of psychological variables. Despite this, the selected solution provides the most coherent and interpretable structure.

5.4.3 Description of Identified Profiles

The clustering procedure divided the 46 users remaining after the data filtering step, where users with a high proportion of missing values were removed. This procedure divided the users into two groups: Profile 1 ($n = 24$) and Profile 2 ($n = 22$), as shown in Figure 19. These profiles were interpreted by computing the mean values of the survey variables for each profile, allowing the identification of characteristic psychological patterns within each group. In Figure 20, Profile 1 is represented in blue, while Profile 2 is represented in orange.

Profile 1 shows higher scores in extraversion, agreeableness, and conscientiousness, together with lower neuroticism. This group also presents higher flourishing and positive affect, and lower levels of loneliness, perceived stress, negative affect, and depressive symptoms. Overall, this profile reflects a psychologically resilient and higher well-being pattern.

Profile 2 presents the opposite tendency, with lower scores in extraversion, agreeableness, and conscientiousness, and higher neuroticism. Users in this group report lower flourishing and positive affect, and higher loneliness, stress, negative affect, and depressive symptoms. This profile represents a more vulnerable well-being pattern.

These profiles should be interpreted as dominant psychological tendencies rather than fixed personality types.

5.4.4 Visualization of Profiles

To support the clustering results, three complementary visualizations were used.

First, the silhouette plot (Figure 19) was used to evaluate the consistency and separation quality of the clusters across different values of k . The maximum silhouette score at $k = 2$ supports the selection of two profiles.

Second, a PCA-based two-dimensional projection (Figure 20) was used to visualize the user distribution in reduced space. Each point represents one user included in the clustering analysis ($n = 46$), positioned according to the first two principal components derived from the standardized survey variables. The projection shows a clear separation trend between the two identified profiles, supporting their structural coherence.

Third, a heatmap of standardized survey means (Figure 21) was constructed to visualize the characteristic psychological patterns of each profile. The heatmap displays the average standardized values of the survey variables within each profile, allowing direct comparison of personality traits and well-being indicators across groups. Positive values indicate scores above the dataset mean, while

negative values represent below-average levels. For example, a positive value in extraversion indicates that a given profile tends to be more extraverted than the overall sample, while a negative value reflects lower-than-average levels. This visualization highlights the contrasting psychological structure of the two profiles and supports their interpretation as distinct well-being patterns.

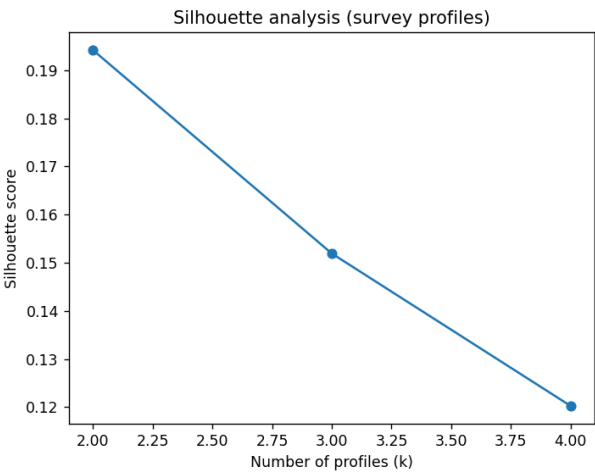


Figure 19. Silhouette Analysis

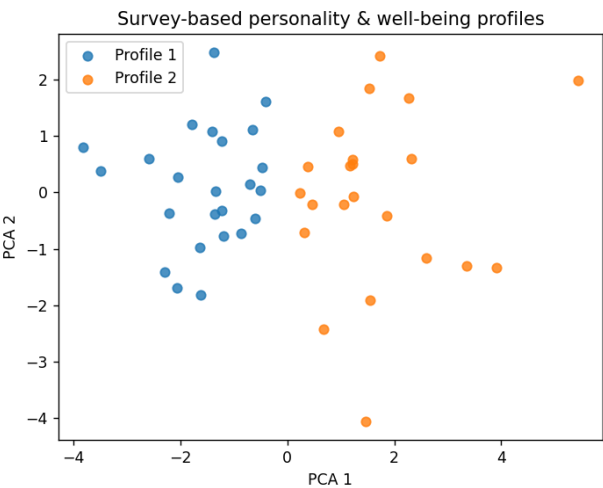


Figure 20. PCA Projection of Profiles

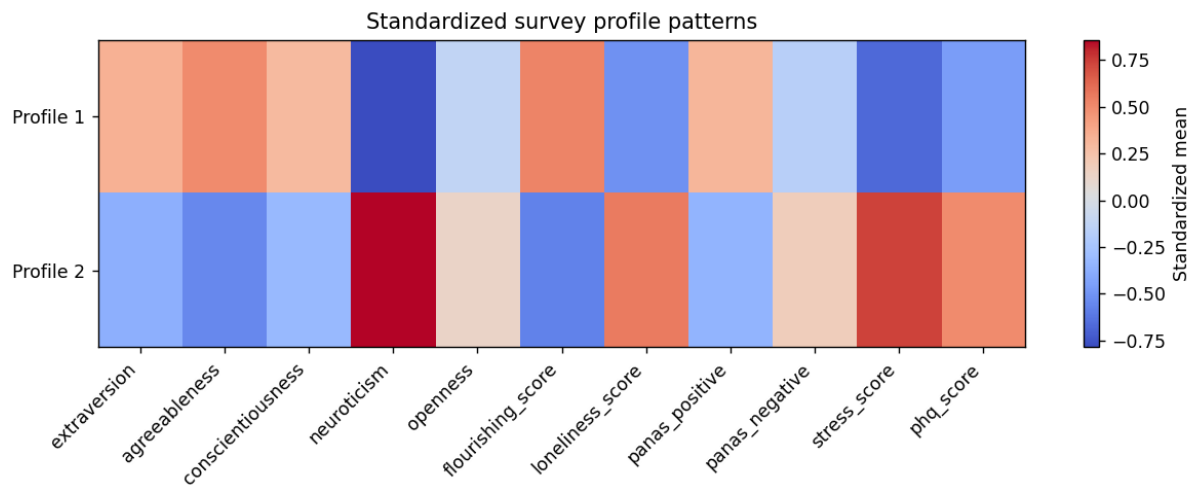


Figure 21. Projection of Profiles

5.4.5 Summary of Survey-Based Profile Identification

Unsupervised clustering on normalized survey measures identified two coherent personality and well-being profiles. The selected solution is supported by silhouette analysis and visualization results. These profiles define the baseline user segmentation used in later stages of the work.

5.5 Comparison of Survey-Based Profiles with EMA and Behavioural Variables

Once the personality and well-being profiles were identified using survey data, the next step was to examine whether these static profiles were associated with differences in users' daily emotional experiences (EMA data) and behavioural patterns. The objective was not to establish causal relationships, but to explore whether individuals with different personality profiles exhibit distinct patterns in self-reported emotions and daily behaviours.

To do this, the previously obtained profile label was merged into the daily dataset, allowing each EMA and behavioural observation to be linked to the corresponding user profile. Only users with a valid identified profile were retained. After merging the profile labels back into the daily dataset, the analysis kept only the daily rows corresponding to those users, resulting in 4,443 daily observations from the same 46 users included in the clustering stage.

5.5.1 Survey Profiles and Daily Emotions

The relationship between personality profiles and daily emotional states was analysed using EMA variables. The selected EMA variables included happiness (happy), general well-being (how), perceived stress (stress_level), and daily social interaction (social_people). These variables are self-reported on a daily basis and reflect subjective emotional and social experiences.

For each profile, the mean value of each EMA variable was computed across all available daily observations. In addition, emotional variability was assessed in two steps. First, the standard deviation of each EMA variable was computed at the user level to capture day-to-day fluctuations in emotional

states. Then, these user-level variability measures were averaged within each profile to allow comparison of emotional stability across groups.

5.5.2 Mean Emotional Levels by Profile

Mean values were computed by grouping all available daily EMA observations according to profile and averaging them across groups. The results indicate that Profile 1 shows slightly higher average levels of positive emotional states (happy and how) compared to Profile 2. Profile 1 also reports, on average, a higher number of daily social interactions. In contrast, Profile 2 presents lower mean happiness and slightly higher perceived stress.

Although the absolute differences are moderate, these patterns are consistent with the personality composition of the profiles identified in the previous section. In particular, Profile 2 is characterized by higher neuroticism and lower scores in positive well-being indicators, which aligns with lower reported happiness and social interaction in daily life.

The boxplot of the happy variable (Figure 22) further illustrates these differences. The visualization is based on 220 daily EMA observations contributed by 38 users, including only days with valid happy reports. While both profiles exhibit overlapping distributions, Profile 1 shows a higher median happiness level and greater variability, whereas Profile 2 displays a greater concentration of lower values, with a single outlier corresponding to an isolated high-happiness day reported by one user.

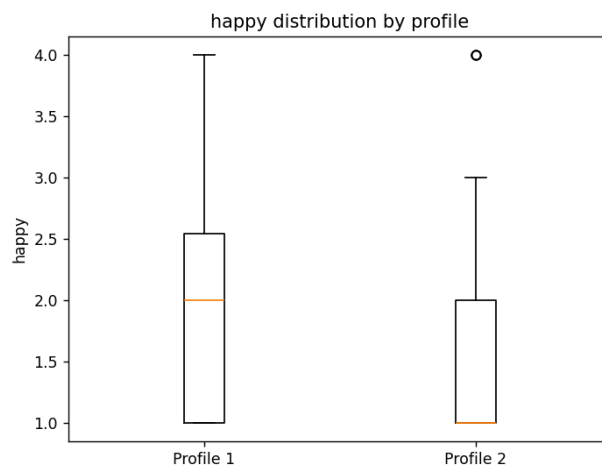


Figure 22. Distribution of Daily Happiness Scores by Profile

5.5.3 Emotional Variability

Emotional variability provides complementary information beyond average levels. The results show that Profile 2 exhibits higher variability in positive emotional variables such as happiness and general well-being, whereas Profile 1 shows slightly higher variability in perceived stress and social interaction. This indicates that the profiles differ not only in average emotional levels, but also in the stability of specific day-to-day experiences. Profile 2 showed more variable daily emotional responses, especially in happiness. This may be partly related to the higher neuroticism levels in this group, since neuroticism is often linked to stronger emotional changes and a greater tendency to experience negative emotions.

Overall, these results suggest that personality profiles derived from survey data are meaningfully reflected in users' day-to-day emotional experiences, even though the analysis remains descriptive.

5.5.4 Behavioural Patterns Across Survey Profiles

In addition to emotional variables, several behavioural indicators were analysed to explore whether personality profiles are associated with differences in daily habits. The selected variables include sleep duration (`sleep_hours`), perceived sleep quality (`sleep_quality`), overall phone activity level (`tiempo_uso_total`), proportion of active time (`prop_active`), and number of applications used (`total_apps_used`). For each profile, mean values of these behavioural variables were computed across all available observations.

The results show that both profiles report similar average sleep duration, with Profile 1 sleeping slightly longer on average. However, Profile 2 reports marginally higher perceived sleep quality despite shorter sleep duration. This highlights that subjective sleep quality does not necessarily align with objective sleep duration. The bar plot of mean sleep hours by profile (Figure 23) visually confirms that sleep duration differences are small. The visualization is based on 1,194 daily behavioural observations aggregated according to the survey-based profiles.

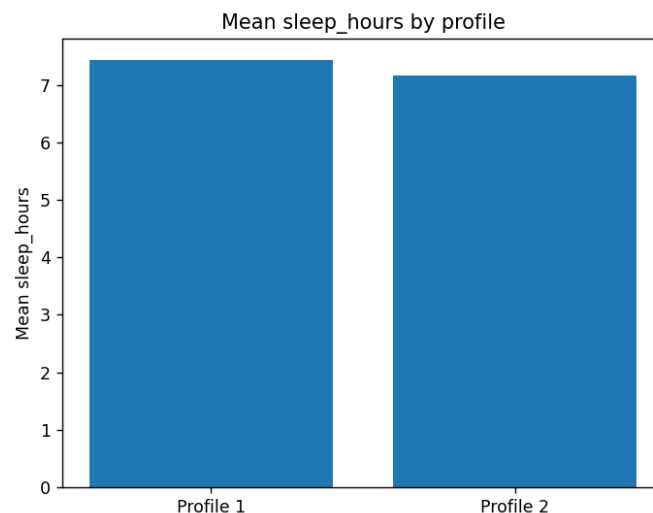


Figure 23. Mean sleep Duration by Profile

More pronounced differences appear in phone usage behaviour. Profile 2 exhibits substantially higher overall device activity and a larger number of applications used, suggesting a more intensive or fragmented interaction with the smartphone. In contrast, Profile 1 shows lower overall phone activity, which may reflect more regulated daily habits.

5.5.5 Summary

Taken together, these analyses indicate that survey-based personality profiles are associated with meaningful, although moderate, differences in daily emotions and behaviours. Profiles characterized by higher well-being and lower neuroticism tend to show higher average happiness, more social

interaction, and more stable emotional patterns. Conversely, profiles associated with higher neuroticism and lower well-being exhibit greater emotional variability and higher levels of overall device activity.

Importantly, this analysis is intentionally descriptive. The goal is not to predict emotions or behaviours from personality alone, but to demonstrate that static personality traits provide a useful baseline for understanding individual differences in daily well-being and behaviour, even when only partial or noisy data are available.

5.6 Personality Profiles and Recommendation Logic

Figure 24 summarizes how the identified personality profiles can be translated into simple, interpretable recommendation strategies. Importantly, recommendations are not derived from predictive models but are grounded in consistent patterns observed across survey, EMA, and behavioural data.

Profile	Key survey characteristics	Observed emotional patterns (EMA)	Observed behavior patterns	Recommendation focus
Profile 1	Higher extraversion, agreeableness, conscientiousness; lower neuroticism; higher flourishing and lower PHQ	Higher average happiness; lower emotional variability; moderate stress levels	Slightly longer sleep duration; lower phone usage; fewer apps used	Maintenance and reinforcement of existing habits; focus on sustaining emotional balance
Profile 2	Lower extraversion and agreeableness; higher neuroticism; lower flourishing; higher stress and PHQ scores	Lower average happiness; higher emotional variability; consistently elevated stress indicators	Slightly shorter sleep duration; higher phone usage; more apps used	Emotional regulation and habit support; focus on stress management and gradual behavioral adjustments

Figure 24. Personality Profiles and Recommendation Logic

5.7 Discussion

5.7.1 Why Survey Data Plays a Key Role

Survey-based information provides a stable and interpretable baseline for well-being analysis. Unlike EMA and behavioural data, which are highly influenced by daily context and temporal fluctuations, survey measures capture relatively stable personality traits and long-term well-being indicators. This stability helps reduce temporal noise and enables meaningful segmentation of users into coherent profiles.

Moreover, survey-based profiles facilitate human-centered interpretation, allowing results to be translated into understandable patterns and actionable insights. In this work, personality traits and well-being scores derived from surveys proved useful for contextualizing emotional and behavioural variability observed at the daily level, supporting their role as a foundational component in personalized well-being analysis.

5.7.2 Limitations

Several limitations should be acknowledged. First, survey data provide a static snapshot of personality and well-being, which may not capture changes over time. Second, survey measures rely on self-reported information, which can be affected by subjective bias. Finally, although survey data provide valuable baseline information, they should be considered complementary to EMA and sensing data rather than a standalone solution. Addressing these limitations opens opportunities for future work that combines dynamic behavioural signals with stable psychological traits.

5.8 Chapter Summary

This chapter presented the use of survey data as a stable baseline for well-being analysis. The relevant psychological instruments were selected and prepared at the user level, including missing data handling and standardization of all variables.

An unsupervised KMeans clustering was then applied to identify personality and well-being profiles, and the resulting solution was evaluated and visualized through silhouette analysis, PCA projection, and a heatmap of standardized means. The identified profiles were then compared with daily EMA and behavioural variables to explore differences across groups, supported by additional visualizations. Finally, the profiles were linked to simple recommendation strategies, and the role and limitations of survey data were discussed.

6 SIMILARITY-BASED CONTEXTUAL RECOMMENDATION

6.1 Motivation and Overview

The previous chapters analysed daily emotional well-being from several complementary perspectives, including exploratory analysis of behavioural patterns, individual variability over time, and personality-based profiles derived from survey data.

These analyses highlighted a key insight: emotional well-being patterns differ substantially across individuals, suggesting that generalized recommendations may fail to capture personal dynamics. For this reason, this chapter moves one step further by introducing a personalized recommendation approach based on contextual similarity.

The core idea is that each individual can learn from their own past experiences: when the current day resembles previous days in terms of context and behaviour, those past days can provide meaningful insight into what has been associated with better emotional outcomes, as reflected by higher happy values.

Accordingly, recommendations are generated by identifying past days of the same user that are most similar to the current day and analysing which contextual variables differed between the current day and similar past days associated with higher happy values.

This approach is consistent with the personalized perspective adopted throughout the project, where within-person comparisons were shown to be more informative than cross-user averages. It also fits the characteristics of the dataset, where emotional EMA observations are sparse and irregular across days.

After preprocessing, the dataset used for the recommendation model contains 220 daily observations with valid happy values from 38 participants, providing the historical behavioural records required for within-person similarity comparisons.

6.2 Day Representation and Feature Standardization

Each day is represented as a vector of 17 contextual and behavioural variables, including sleep patterns, social interaction, smartphone usage, activity levels, and academic workload. This representation allows daily records to be structured in a comparable format suitable for similarity computation.

The selection of variables follows the dimensions previously analysed in the project, where emotional well-being was examined in relation to daily behavioural and contextual factors. By combining sensing indicators and self-reported measures, each day captures a multidimensional view of the individual's daily context.

The feature set includes 17 contextual indicators capturing multiple aspects of daily behaviour: sleep patterns (sleep_hours, sleep_quality), psychological state (stress_level), social interaction

(social_people), smartphone activity (tiempo_uso_total, total_apps_used, locked_minutes, dark_minutes), mobility and activity patterns (prop_active, prop_stationary), digital environment indicators (unique_wifi, bt_unique_devices), communication behaviour (total_calls, num_sms), and academic workload (calendar_event_count, num_deadlines_today, hours). Since these variables are measured on different scales and may present very different typical values across individuals, feature standardization is required. Without normalization, variables with larger numeric ranges would dominate similarity calculations, introducing distortions in the comparison process.

For this analysis, the original dataset is restricted to the selected contextual features and to observations with valid happiness values, ensuring that similarity comparisons are based on consistent and relevant information.

To ensure that similarity comparisons reflect deviations from an individual's usual behaviour, all features are standardized within each user using z-score normalization:

$$z = \frac{x - \mu_{user}}{\sigma_{user}}$$

Equation 1. Within-user z-score normalization

where x is the observed value for a given day, and μ_{user} and σ_{user} correspond to the mean and standard deviation of that variable for the same user.

This within-user standardization ensures that similarity reflects whether a value is high or low for that individual, rather than being influenced by differences across users.

6.3 Similarity Computation Between Days

For a given user and a given day t , the recommendation system compares the current day only with previous days of the same user, avoiding the use of any future information. This design preserves the temporal logic of the data and ensures that recommendations are based exclusively on past observed experiences.

Each day is represented as a standardized feature vector, as described in the previous section. Let x_t denote the vector corresponding to the current day, and x_i the vector corresponding to a past day of the same user.

Similarity between days is computed using Euclidean distance, defined as:

$$d(x_t, x_i) = \sqrt{\sum_j (x_{i,j} - x_{t,j})^2}$$

Equation 2. Euclidean distance between two days

This metric quantifies the overall contextual difference between two days by aggregating the squared differences across all standardized features. Smaller distances indicate higher similarity, while larger distances reflect more dissimilar daily contexts.

Euclidean distance was selected due to its interpretability and suitability for standardized continuous variables. Since all features were normalized within each user, the metric captures relative behavioural deviations consistently across dimensions without being dominated by variables with larger scales.

To ensure that similarity can be computed for all days, missing values were replaced with zero after feature standardization. Since features are normalized using z-scores, a value of zero represents a typical or average behaviour for the user. In practice, this means that when a value is missing (for example, `sleep_hours`), it is treated as if the user had a typical value for that variable on that day. This simplifies the computation, as it allows distances to be calculated without handling missing values explicitly and avoids discarding observations. However, this assumption may reduce the ability of the model to distinguish between days, since missing values are treated as normal behaviour rather than unknown information.

After computing distances between the current day and all past days of the same user, only the K most similar days (nearest neighbours) are retained for subsequent recommendation analysis. This selection focuses the comparison on the most contextually relevant past experiences. In this implementation, K is set to 10. This value provides a balance between capturing sufficiently similar contextual situations while maintaining a diverse set of historical examples for comparison.

6.4 Identification of Better-Than-Usual Days

Not all similar days are equally informative for recommendation purposes. To focus on emotionally relevant experiences, days are classified as better-than-usual based on the user's own emotional baseline.

A day is labelled as better-than-usual if its reported happiness score exceeds the user's average happiness across the observation period. This within-person definition avoids direct emotional comparisons across individuals and remains consistent with the personalized analytical perspective adopted throughout the project.

Among the K most similar past days identified in the previous step, only those classified as better-than-usual are retained for recommendation derivation. This filtering ensures that recommendations are grounded in contexts where the user experienced relatively improved emotional outcomes. In cases where none of the K nearest neighbours are classified as better-than-usual, the system is unable to derive recommendations from emotionally positive reference days.

6.5 Personalised Recommendations Derivation

For the subset of similar days associated with higher happy values (i.e., better-than-usual days for the user), the system computes the average of the contextual variables and compares these values with

those of the current day. This comparison identifies which variables tend to be higher or lower on similar days with higher happy values relative to the user's baseline.

The resulting differences are interpreted as potential behavioural or contextual adjustments associated with improved well-being. To maintain interpretability, only the variables with the largest absolute differences are retained, producing a concise and meaningful set of personalized recommendations. Absolute differences are used to identify the contextual variables that most strongly distinguish emotionally better days from the current one, regardless of direction.

The overall process for identifying similar days and deriving recommendations is summarized in Figure 25.

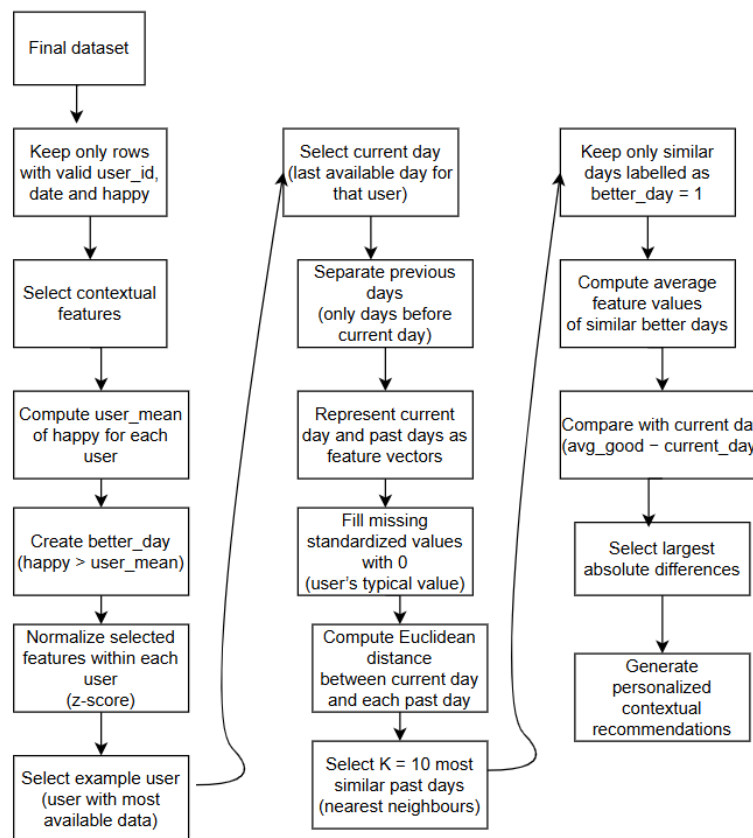


Figure 25. Similarity-Based Recommendation Workflow

6.6 Example Results and Interpretation

An illustrative example is presented for User 59 on 2013-06-07, a day with a reported happiness score of 2.0, indicating a relatively low emotional state for this individual. For this user, a total of 42 historical days were available for comparison prior to the target date.

To identify meaningful contextual references, the system retrieves the K most similar past days based on the similarity computation described in previous sections.

Most similar past days

Among the 10 nearest neighbours, 4 days correspond to above-average emotional states for that user and are therefore labelled as better-than-usual. These days constitute the reference subset used to derive personalized recommendations.

For clarity, only the top five most similar days are displayed in Figure 2. Each row represents a past day, showing the date, the distance to the current day (where lower values indicate higher similarity), the observed happiness score, and whether the day was classified as better-than-usual. The rows are ordered by increasing distance, so that the most similar days appear first.

These similar past days share contextual and behavioural patterns with the current day, making them suitable comparison points.

date	dist	happy	better_day
2013-06-02	2.138734	3.250000	1
2013-06-06	2.483560	2.000000	0
2013-06-03	2.567234	3.333333	1
2013-05-23	2.912374	3.500000	1
2013-06-01	3.606615	2.000000	0

Figure 26. Most Similar Past Days for the Selected User

Derived recommendations

By comparing the values of the current day with the average values of similar better days, the system identifies the variables that differ the most. To maintain interpretability, only the six variables with the largest absolute differences are reported, providing a concise set of personalized recommendations.

In this example, the comparison yields the following patterns:

- Higher overall device activity (*tiempo_uso_total*: +1.05)
- Longer sleep duration (*sleep_hours*: +0.88)
- More social interaction (*social_people*: +0.77)
- Lower academic workload (*hours*: -0.73)
- More stationary time (*prop_stationary*: +0.60)
- Lower physical activity (*prop_active*: -0.52)

Interpretation

For this user, emotionally better days under similar contextual conditions tend to involve more sleep, greater social interaction, and higher levels of overall device activity, while being less physically active and academically less demanding.

Importantly, these patterns are user-specific and should not be interpreted as universal behavioural recommendations, but rather as personalized insights derived from the individual's own historical patterns. This example illustrates how similarity-based contextual analysis can transform historical behavioural patterns into interpretable and personalized well-being recommendations.

6.7 Discussion

This similarity-based approach does not aim to predict emotions or establish causal relationships. Instead, it functions as a reflective tool that helps individuals understand which contextual patterns have historically been associated with better emotional outcomes in their own lives.

Importantly, the main contribution of the proposed system lies in its interpretability and personalization. By identifying contextual and behavioural differences between the current day and emotionally better past experiences, the system supports self-awareness and reflection grounded in individual context.

Overall, the approach aligns with the goals of human-centered and responsible digital well-being technologies, positioning similarity-based reflection as a valuable complement to traditional predictive systems.

One limitation of this approach is that recommendations are generated regardless of the current emotional state. Even when the current day already presents higher-than-usual happy values, the system still compares it with similar better-than-usual days. As a result, the recommendations may not necessarily aim at improving the current state, but rather reflect patterns consistently observed in past days associated with higher happy levels, highlighting contextual conditions that tend to co-occur with more positive emotional states.

However, this also represents a strength of the approach, as the system can provide insight into the user's own behavioural and contextual patterns, helping to understand which factors are typically present in more favourable emotional situations. In this sense, the method should be interpreted as a personalized exploratory tool rather than a strictly prescriptive recommendation system.

6.8 Complementary Rule-Based Recommendation Analysis

6.8.1 Justification

While the similarity-based recommendation approach provides a personalized way of identifying behavioural patterns associated with improved emotional outcomes, it also focuses on a very specific perspective. In particular, the method is designed to analyse the current day by comparing it with similar past situations, which means that the recommendations are always linked to a single moment in time.

As discussed in the previous section, this approach follows a within-user perspective and allows the analysis to be based on the individual's own experience. However, because it is centred on one specific

day, it mainly captures patterns related to that situation, instead of showing more general behavioural tendencies across the user's history. For this reason, it is useful to extend the analysis by incorporating a broader perspective.

To do so, a complementary rule-based approach is introduced. Instead of focusing on similarity between individual days, this method aims to identify patterns that appear multiple times in the user's data and are associated with better-than-usual emotional states.

By looking at these patterns across several days, the rule-based approach allows the analysis to move from specific situations to more general behavioural trends within the same user. This makes it possible to identify consistent relationships between contextual factors and emotional outcomes, and not only comparisons with a single reference day.

Importantly, this new approach is not intended to replace the similarity-based method, but to complement it. The similarity-based model focuses on situations that resemble the current day, while the rule-based approach captures patterns that appear across the user's history and are linked to better emotional states. Together, both approaches provide a more complete understanding of personalized well-being, combining detailed analysis of individual situations with a more general view of behaviour.

6.8.2 Methodological Approach and Implementation

6.8.2.1 Data Preparation and Preprocessing

The rule-based analysis is applied to the preprocessed dataset, which is filtered to retain only observations with valid values for the target variable (happy), resulting in 220 daily observations from 38 users.

As in the previous approach, the variable `better_day` is used to identify days where the user's happiness is above their personal average. In addition, all selected features are normalized within each user using z-score standardization, so that each value reflects whether it is high or low relative to that individual's usual behaviour.

6.8.2.2 Feature Discretization

After normalization, the continuous z-score values are transformed into categorical levels to simplify the rule extraction process. Each feature is discretized into three categories: *LOW*, *NORMAL*, and *HIGH*. Values below -0.5 are labelled as *LOW*, values above 0.5 as *HIGH*, and values in between as *NORMAL*, while missing values remain undefined. This transformation allows the model to work with discrete states instead of continuous values.

6.8.2.3 Selection of Contextual Variables

For this analysis, a subset of contextual variables is selected to represent different aspects of daily life related to emotional well-being. These include the perceived stress level (`stress_level`), overall device activity (`tiempo_uso_total`) and physical activity patterns through the proportion of active time

(prop_active) and stationary time (prop_stationary). Social interaction is captured through communication variables such as the number of calls (total_calls) and messages (num_sms), while mobility and environmental variation are represented using the number of unique WiFi networks detected (unique_wifi). In addition, daily structure and workload are reflected through the number of scheduled events (calendar_event_count) and of academic deadlines (num_deadlines_today). Together, these variables provide a concise description of the user's daily context, suitable for identifying patterns in the rule-based approach.

6.8.2.4 Rule Extraction Parameters

To extract rules, three parameters are defined:

- A minimum support of 4 means that a condition must appear in at least four days of the user's data to be considered relevant.
- A minimum confidence of 0.60 means that, among the days where the condition appears, at least 60% must correspond to better days (happy).
- The maximum rule size is limited to two variables, so that the patterns remain simple and easy to interpret.

6.8.2.5 Rule Extraction Procedure

To extract the rules, a function is applied separately to each user. This function uses the categorical variables defined in the previous step and searches for patterns associated with better days.

First, the method identifies all categorical features, which represent the transformed version of the original variables. It then evaluates simple conditions based on a single feature. For each categorical feature, the method first extracts the set of unique, non-missing values (*LOW*, *NORMAL*, or *HIGH*). For each of these values, it selects the days where the condition is satisfied and computes how often it appears (support) and how often it is associated with better days (confidence). Only the conditions that meet the minimum thresholds are retained as valid rules.

Next, the method extends the same procedure to combinations of two variables. It evaluates pairs of conditions (for example, *stress_level* = *LOW* and *num_sms* = *HIGH*) and applies the same filtering criteria based on support and confidence. All possible pairs of variables are generated without considering duplicate pairs (e.g., (A, B) and (B, A)), ensuring that each combination is evaluated only once. In addition, rows with missing values are excluded so that rules are built only from complete observations. This allows the identification of simple interactions between variables while maintaining interpretability.

Finally, all the extracted rules are collected and ordered according to their confidence and support, prioritizing those that are more strongly associated with better days and appear more frequently.

Overall, the process used to extract rule-based patterns is summarized in Figure 27.

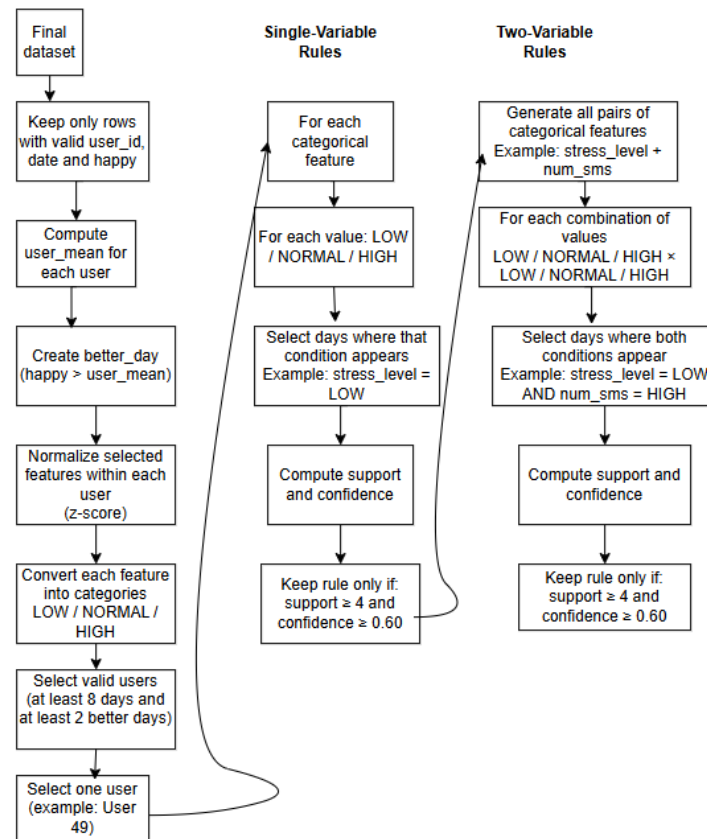


Figure 27. Rule-Based Within-User Pattern Extraction Workflow

6.8.3 Results and Interpretation

To illustrate the results, several users with enough available data are selected. The analysis is applied individually to each user, since the rules are extracted within each person.

For example, for user 59, a total of 43 daily observations with valid happy values are available, and the proportion of better days is 0.349. This means that approximately 34.9% of the observed days correspond to above-average emotional states for this user. As shown in Figure 28, the extracted rules represent recurring patterns associated with better-than-usual days for the selected user. The rules are filtered based on minimum thresholds and primarily ordered by confidence. Only the top 10 rules are displayed.

```

Selected user: 59
Days available: 43
Better-day rate: 0.349

Top within-user rules associated with better days:

```

rule	support	confidence
calendar_event_count=HIGH	4	0.750
prop_stationary=LOW AND unique_wifi=HIGH	4	0.750
prop_stationary=LOW AND total_calls=NORMAL	4	0.750
calendar_event_count=HIGH AND num_deadlines_today=LOW	4	0.750
stress_level=HIGH AND tiempo_uso_total=HIGH	7	0.714
unique_wifi=HIGH AND total_calls=NORMAL	9	0.667
tiempo_uso_total=HIGH AND num_sms=LOW	6	0.667
stress_level=HIGH AND tiempo_uso_total=LOW	8	0.625
stress_level=HIGH AND unique_wifi=LOW	8	0.625
stress_level=HIGH AND num_deadlines_today=LOW	15	0.600

Figure 28. Rule-Based Within-User Pattern Extraction Workflow

For this user, the results suggest that better days are often associated with specific combinations of daily conditions rather than a single factor. Patterns involving stress level, device activity, and daily workload appear frequently among the top rules. This indicates that the user's emotional state is influenced by a combination of behavioural and contextual factors, and that certain configurations of these variables are more likely to be linked to improved well-being.

A second example is shown for user 49, who has 16 available days with happy values and a better-day rate of 0.438. In this case, the extracted rules suggest that better days are often associated with lower stress levels and moderate or balanced activity patterns for this user. In particular, combinations involving normal communication levels, moderate movement, and controlled device activity appear frequently. This indicates that, for this user, maintaining a balanced routine rather than extreme behaviours is more likely to be linked to improved emotional states. As shown in Figure 29, similar rule-based patterns are observed for this user, with the top 10 rules displayed.

```

Selected user: 49
Days available: 16
Better-day rate: 0.438

Top within-user rules associated with better days:

```

rule	support	confidence
stress_level=LOW	4	0.750
tiempo_uso_total=HIGH AND prop_stationary=NORMAL	4	0.750
prop_active=LOW AND total_calls=NORMAL	4	0.750
prop_active=LOW AND num_sms=NORMAL	4	0.750
prop_stationary=HIGH AND unique_wifi=NORMAL	4	0.750
tiempo_uso_total=HIGH	7	0.714
tiempo_uso_total=HIGH AND calendar_event_count=NORMAL	6	0.667
tiempo_uso_total=HIGH AND unique_wifi=NORMAL	5	0.600
tiempo_uso_total=HIGH AND total_calls=NORMAL	5	0.600
tiempo_uso_total=HIGH AND num_sms=NORMAL	5	0.600

Figure 29. Rule-Based Recommendations for User 49

6.9 Chapter Summary

This chapter proposed a personalized recommendation approach based on contextual similarity, allowing each day to be analysed in relation to similar past experiences of the same user. This method provides insights linked to specific situations and helps identify which contextual factors are associated with better emotional outcomes.

To extend this perspective, a rule-based approach was also introduced. Unlike the similarity-based method, this approach identifies general patterns across the user's history, highlighting conditions that are consistently associated with better days.

The results show that emotional well-being depends on a combination of factors and that these patterns differ between users. Overall, combining both approaches provides a more complete understanding of individual behaviour, capturing both specific situations and more general trends.

7 IDENTIFYING BEHAVIOURAL AND CONTEXTUAL FACTORS ASSOCIATED WITH BETTER-THAN-USUAL DAYS

Figure 30 summarizes the main steps of this chapter.

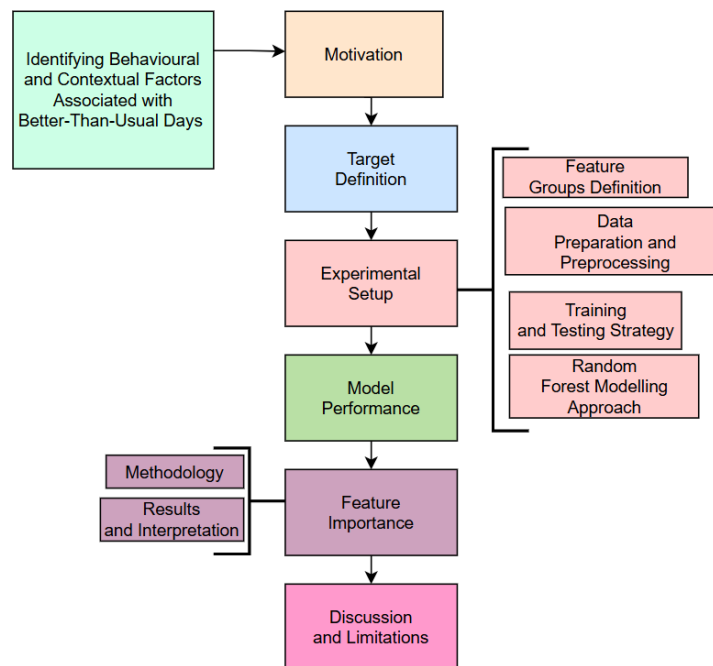


Figure 30. Analytical Workflow for Identifying Behavioural and Contextual Factors Associated with better-than-usual Days

7.1 Motivation

Previous analyses examined daily emotional well-being from a within-person perspective. In particular, daily happiness values were evaluated relative to each participant's own average level, allowing days to be classified as better-than-usual or at-or-below-usual. This classification allows daily emotional states to be analysed relative to each participant's own average level rather than through direct comparisons across users.

Using this distinction, contextual and behavioural variables were compared across the two types of days in order to explore how daily conditions relate to variations in reported happiness. This descriptive comparison provided an initial view of how daily context may differ between relatively better and worse emotional days.

While these analyses focused on descriptive comparisons, they raise an additional analytical question: *whether the contextual variables available in the dataset contain enough information to distinguish better-than-usual emotional days from other days.*

To explore this question, the present chapter examines how different groups of contextual variables relate to the identification of better-than-usual emotional days (i.e., days where happy exceeds the participant's own average). The objective is not to develop a predictive system, but rather to analyse whether daily contextual information provides detectable signals associated with whether a day is better-than-usual, where this is defined relative to each participant's own average emotional level. In particular, the aim is to examine whether such patterns are consistent across individuals, for example, not that everyone behaves the same way, but that when a person behaves differently from their own usual pattern (e.g., being more active than usual), this may be associated with better emotional days in a consistent way across users.

7.2 Target Variable Definition

The analysis relies on the distinction between better-than-usual days introduced in the previous exploratory analysis. In that analysis, daily happiness values were evaluated relative to each participant's own average happiness level.

Following the same approach, a binary outcome variable called `better_day` is defined. For each user, the mean happiness value across their available days is computed and used as a personal reference level. A day is labelled as `better_day = 1` if its happiness value is higher than the participant's average happiness, and `better_day = 0` otherwise.

After filtering observations with valid happiness reports, the dataset used in this analysis contains 220 daily observations from 38 users. Applying the above definition results in 72 better-than-usual days and 148 remaining days. Although the dataset contains a limited number of observations, both categories remain sufficiently represented to explore whether contextual variables contain detectable signals associated with better-than-usual emotional states.

7.3 Experimental Setup

To examine whether contextual variables contain detectable signals associated with better-than-usual emotional days, an experimental modelling framework was implemented using the contextual variables available in the dataset. The purpose of this approach is to analyse how different groups of daily contextual variables relate to the identification of better-than-usual emotional states.

7.3.1 Definition of Feature Groups

The contextual variables used in the analysis were organized into several feature groups representing different dimensions of daily behaviour and context. This grouping allows the analysis to examine which types of contextual information appear more informative for distinguishing better-than-usual days.

The following groups were considered:

- Academic context: variables related to academic workload and daily schedule (calendar events, deadlines, and hours of activity).
- Sleep-related factors: indicators describing sleep duration and perceived sleep quality.
- Psychological factor: perceived stress level reported in daily EMA responses.
- Social interaction: variables reflecting communication and social exposure, including number of people interacted with, nearby Bluetooth devices, calls, and SMS activity.
- Phone behaviour: indicators of smartphone usage such as overall device activity, number of applications used, screen inactivity periods, and device locking time.
- Mobility and activity: variables derived from sensing data describing movement patterns and environmental context, including activity proportions and number of detected Wi-Fi networks.

In addition to evaluating these groups separately, a final configuration including all available contextual variables was also analysed. This allows comparing whether combining multiple behavioural dimensions improves the ability to distinguish better-than-usual days.

7.3.2 Data Preparation and Preprocessing

Before training the models, several preprocessing steps were applied to ensure that the contextual variables could be analysed consistently.

First, all contextual variables were converted to numeric format to ensure compatibility with the modelling process.

In addition, contextual variables were normalized within each user using z-score standardization. This transformation expresses each value relative to the participant's usual behaviour, allowing the model to analyse whether higher or lower values than usual are associated with better-than-usual happiness days. For example, after z-score standardization, sleeping 8 hours may be above usual for one participant but below usual for another.

7.3.3 Training and Testing Strategy

To evaluate whether contextual variables contain information associated with better-than-usual emotional days, the data were divided into training and testing subsets. However, since the dataset contains multiple daily observations for the same participant, a standard random split at the row level would not be appropriate. Such a split could place days from the same user in both the training and testing sets, allowing the model to benefit from user-specific patterns already seen during training.

To avoid this issue, the split was performed at the user level using a group-based strategy (implemented using `GroupShuffleSplit`). In this way, all observations from a given participant were assigned either to the training set or to the testing set, but never to both simultaneously. This design allows the evaluation to better reflect whether the model captures general contextual patterns rather than simply memorizing characteristics of specific users.

Following this strategy, approximately 80% of the users were assigned to the training set and the remaining 20% to the test set. In the implemented split, the training set contained 166 daily observations from 30 users, while the test set contained 54 daily observations from 8 users.

Missing values were handled within the modelling pipeline using median imputation. For each contextual variable, the median value was calculated using only the training data and was then used to replace missing values in both the training and test sets. This ensures that no information from the test set is used during training, preventing potential information leakage and maintaining a valid evaluation of the model's performance.

7.3.4 Random Forest Modelling Approach

To examine whether contextual variables contain information related to better-than-usual emotional days, a Random Forest classifier was applied. This model was chosen because it can capture nonlinear relationships between variables and flexibly combine multiple contextual factors without requiring strong assumptions about how these variables relate to emotional outcomes. In addition, Random Forest models can perform robustly with relatively limited datasets and diverse types of variables, which makes them suitable for the contextual features available in this study.

The model was implemented using 300 decision trees, with a maximum depth of 8 and a minimum of 5 observations per leaf. In addition, class balancing was applied to account for the unequal distribution between better-than-usual days and the remaining days in the dataset.

The same modelling procedure was applied to each contextual feature group defined earlier, as well as to the full set of contextual variables. This allows the analysis to examine how different contextual dimensions relate to the identification of better-than-usual emotional days.

7.4 Model Performance

The objective of this section is to examine how different contextual dimensions relate to the identification of better-than-usual emotional days. To do so, separate models were trained for each feature group, allowing a structured comparison of their contribution. Model performance was primarily assessed using ROC-AUC, since this metric captures how well the model distinguishes between better-than-usual and typical days based on predicted probabilities. This is particularly useful in the present setting, where class distributions are not perfectly balanced and accuracy alone may overstate performance.

In addition, complementary evaluation metrics such as accuracy, the confusion matrix, and the classification report were examined to provide a more detailed understanding of model performance across classes.

Among the individual feature groups, academic context showed the strongest overall signal, with a ROC-AUC of 0.648. This suggests that academic-related variables contain meaningful information about daily emotional variation. However, the confusion matrix and classification report show that the model performs better in identifying non-better days than better-than-usual days, indicating that academic context alone does not fully capture positive emotional variations. This means that, although the model can assign relatively higher probabilities to better-than-usual days than to non-better days, these probabilities are often not high enough to classify them as positive cases. As a result, several better-than-usual days may still be labelled as non-better days.

Sleep-related variables also showed relatively strong performance, with a ROC-AUC of 0.630. In this case, the model achieved a more balanced behaviour across classes, showing a better ability to detect better-than-usual days than the academic model. Even so, the overall results remained moderate, suggesting that sleep contributes relevant information but is not sufficient on its own to reliably identify emotionally better days.

By contrast, the psychological feature group, represented only by stress level, showed very limited performance, with a ROC-AUC of 0.517. This result is close to random discrimination and indicates that stress level alone provides little useful information for distinguishing between better-than-usual and typical days. A similar pattern appears in the social interaction group, which also produced low performance (ROC-AUC = 0.549), suggesting that social variables in isolation do not provide a strong signal for identifying positive emotional deviations.

Phone behaviour produced intermediate results, with a ROC-AUC of 0.599. Although this performance is still moderate, the model showed a better balance across classes and a clearer improvement in the detection of better-than-usual days compared with several previous feature groups. This suggests that digital behaviour may capture part of the context associated with emotionally better days, although not strongly enough to support reliable identification on its own.

Mobility and activity variables showed relatively weak performance (ROC-AUC = 0.540), again with lower effectiveness in detecting better-than-usual days. This indicates that these variables, when considered separately, provide only limited information about positive emotional variation.

Finally, the model including all contextual variables achieved the highest accuracy (0.667) and a ROC-AUC of 0.612. This suggests that combining multiple dimensions provides a broader representation of daily context and improves overall classification performance. However, the model still showed greater success in identifying non-better days than better-than-usual days. Taken together, these results indicate that emotionally better days are difficult to capture from any single contextual dimension and remain challenging even when multiple behavioural signals are combined.

7.5 Feature Importance

7.5.1 Methodology

After comparing the performance of the different feature-group models, the next step is to analyse the role of the individual variables included in each model. The previous section focused on comparing groups of contextual features. In this section, the analysis moves to the variable level to identify which specific features contribute most within each trained model.

For each feature group, feature importance was estimated after the Random Forest model had already been trained and evaluated on the corresponding train-test split. To compute this importance, the test data were first passed through the imputation step of the pipeline, where missing values were replaced using the median values estimated from the training data. This was necessary because the model had been trained on data in which missing values were replaced using median imputation, so the same

preprocessing had to be applied before analysing variable importance. Once this step was completed, the trained Random Forest classifier was extracted from the pipeline.

Feature importance was then computed using permutation importance. This method evaluates one variable at a time. For a given feature, its values in the test set are randomly shuffled, while all other variables are left unchanged. In this way, the relationship between that feature and the target is intentionally broken. The model is then evaluated again to see how much performance worsens. If performance drops substantially, the feature was contributing useful information to the model. If performance changes very little, the feature has only a minor role. In some cases, the value may even be slightly negative, which suggests that the variable adds little useful signal or may introduce noise in that specific model. For each feature group, a model was trained, and the importance of its variables was then evaluated using permutation importance.

To ensure consistency with the model comparison presented in the previous section, the effect of each permutation was measured using ROC-AUC. This means that feature importance is interpreted as the average decrease in ROC-AUC produced when a given variable is randomly shuffled. In this implementation, the permutation process was repeated 20 times for each feature ($n_repeats = 20$) in order to reduce random variation and obtain a more stable estimate. The final importance value corresponds to the mean decrease in ROC-AUC across those 20 repetitions.

Finally, the resulting importance values were stored in a table containing the name of each feature and its average importance. These values were then sorted from highest to lowest, allowing the most influential variables in each model to be identified more easily.

7.5.2 Results and Interpretation

Within the academic context model, calendar-related variables clearly dominate. In particular, `calendar_event_count` shows a substantially higher importance than the rest, indicating that daily academic structure plays a relevant role in explaining emotional variation. However, the remaining variables contribute only marginally, suggesting that the predictive signal in this group is concentrated in a single feature rather than distributed across all variables. Table 10 shows the permutation importance values for the academic context model, indicating how much each variable contributed to the model's ROC-AUC performance.

Table 10. Feature Importance in the Academic Context Model

Feature	Importance Mean
<code>calendar_event_count</code>	0.118928
<code>num_deadlines_today</code>	0.006854
<code>hours</code>	0.005753

In the sleep-related model, `sleep_quality` emerges as the most influential variable, with a notably higher importance than `sleep_hours`. This suggests that subjective sleep perception is more

informative than sleep duration when identifying better-than-usual days. Moreover, both variables contribute positively, reinforcing the idea that sleep-related factors provide a consistent signal. Table 11 presents the feature importance values for the sleep-related model, showing that sleep_quality contributed more to the model's performance than sleep_hours.

Table 11. Feature Importance in the Sleep-Related Model

Feature	Importance Mean
sleep_quality	0.131605
sleep_hours	0.054972

The psychological model, based solely on stress_level, shows a limited contribution. Although the variable has a positive importance, its magnitude is relatively small, which is consistent with the low ROC-AUC observed previously. As shown in Table 12, stress_level had a positive but relatively low importance value. This suggests that stress alone is not sufficient to capture improvements in emotional states.

Table 12. Feature Importance in the Stress Model

Feature	Importance Mean
stress_level	0.035973

The social interaction group shows generally low importance values, with one variable (num_sms) even exhibiting negative importance. As shown in Table 13, most social interaction variables contributed only weakly to the model, while num_sms showed a negative contribution. This suggests that, within this model, some variables may introduce noise or redundant information rather than improving predictive performance.

Table 13. Feature Importance in the Social Interaction Model

Feature	Importance Mean
social_people	0.046449
bt_unique_devices	0.018075
total_calls	0.015163
num_sms	-0.067720

The phone behaviour model presents a more balanced distribution of importance across variables. As shown in Table 14, total_apps_used and dark_minutes were the most influential variables, suggesting

that patterns of device usage may capture relevant aspects of daily behaviour linked to emotional states.

Table 14. Feature Importance in the Phone Behaviour Model

Feature	Importance Mean
total_apps_used	0.057457
dark_minutes	0.056818
tiempo_uso_total	0.033097
locked_minutes	0.014986

In the mobility and activity group, variables such as unique_wifi and prop_stationary display moderate importance, although overall contributions remain limited. This aligns with the relatively low performance of this group in distinguishing better-than-usual days. Table 15 shows these feature importance values.

Table 15. Feature Importance in the Mobility and Activity Model

Feature	Importance Mean
unique_wifi	0.046023
prop_stationary	0.045597
prop_active	0.015696

When considering the model trained with all features, the importance ranking changes notably. Variables such as sleep_quality and stress_level appear as the most influential, while others that were important in individual models, such as calendar_event_count, show a reduced contribution. This reflects the fact that feature importance is context-dependent: when multiple related variables are combined, some features may become less relevant due to redundancy or shared information.

In addition, several variables exhibit near-zero or negative importance values in the full model. This suggests that, once all contextual dimensions are considered together, certain features do not provide additional useful information and may even introduce noise. This may occur when their relationship with better-than-usual days is weak, inconsistent across users, or already captured by other related variables. Table 16 shows these feature importance values.

Table 16. Feature Importance Using All Features

Feature	Importance Mean
sleep_quality	0.023153
stress_level	0.018111
calendar_event_count	0.012145
unique_wifi	0.006818
hours	0.000994
num_deadlines_today	0.000710
social_people	-0.001065
dark_minutes	-0.001207
prop_stationary	-0.001918
sleep_hours	-0.004545
prop_active	-0.005256
locked_minutes	-0.006960
total_apps_used	-0.009943
tiempo_uso_total	-0.010156
bt_unique_devices	-0.010582
total_calls	-0.011009
num_sms	-0.012145

Figure 31 presents the permutation importance values for the model trained using all features. The plot highlights the relative contribution of each variable in terms of the decrease in ROC-AUC when permuted.

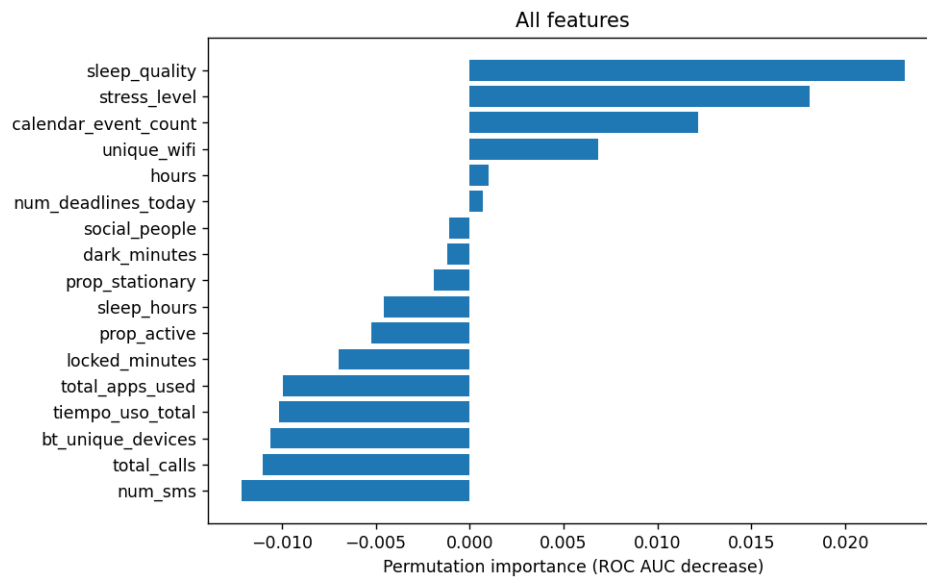


Figure 31. Feature Importance in the All-Features Model

Overall, these results confirm that emotional variation is influenced by multiple interacting factors. While some variables consistently contribute more strongly, no single feature dominates across all models, highlighting the complexity of predicting better-than-usual emotional days.

The complete modelling pipeline, from data preprocessing to performance evaluation and feature importance analysis, is summarized in Figure 32.

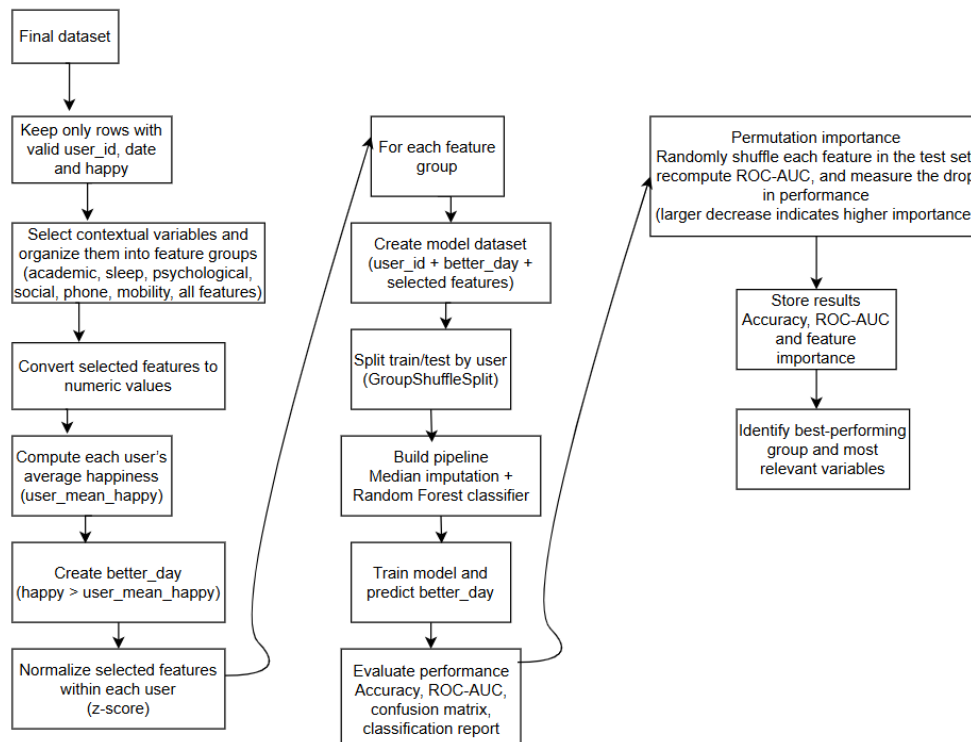


Figure 32. Modelling and Feature Importance Workflow

7.6 Discussion and Limitations

The results show that predicting better-than-usual emotional days is a challenging task, with only moderate performance across all models. While some feature groups, such as academic context and sleep-related factors, provide useful information, no model can reliably identify positive emotional deviations.

A consistent pattern across all models is the better performance in identifying typical (non-better) days compared to better-than-usual ones. This suggests that positive emotional states are harder to capture, likely because they are less frequent and depend on a combination of factors that are not always fully reflected in the data.

The permutation importance analysis supports this idea. Most variables show relatively small importance values, indicating that no single feature strongly determines the predictions. Instead, variations in daily happiness seem to be influenced by multiple factors, each contributing only a small part.

These results are also affected by limitations in the dataset. The sample size is relatively small and uneven across users, which makes it difficult for the model to learn consistent patterns. In addition, several contextual variables contain missing values, which were handled through median imputation within the modelling pipeline. While this approach allows the model to operate on incomplete data, it may reduce variability and weaken the contribution of variables with a high proportion of missing observations. Furthermore, the target variable (“better_day”) is defined relative to each user’s average mood. This means that a day is considered “better” only in comparison to that person’s usual state, rather than based on a fixed threshold. As a result, small variations can change the class label, making the distinction between classes less clear and the prediction task more difficult.

Overall, the results suggest that, although the available features capture part of the context related to emotional variation, they do not fully explain when a day will be better than usual. This highlights the complexity of emotional states and the need for richer or more detailed data to improve prediction.

At the same time, it is important to consider the nature of this problem, as emotional well-being is inherently personal and context dependent. The use of a relative target such as “better_day” makes the analysis more meaningful at an individual level, but also increases the difficulty of the prediction task, since the classification is not based on a fixed threshold. As a result, similar values can correspond to different classes depending on the user, making the distinction between classes less clear.

7.7 Chapter Summary

This section analysed how different contextual feature groups relate to the identification of better-than-usual emotional days. The results show that some groups, particularly academic context and sleep-related factors, provide more useful information than others, although overall performance remains moderate.

Across all models, a consistent pattern appears: the models are more effective at identifying typical days than better-than-usual days. This reflects the difficulty of capturing positive emotional deviations, which are less frequent and more dependent on complex combinations of factors.

The feature importance analysis further supports this idea, showing that no single variable dominates the predictions. Instead, multiple features contribute small amounts of information, highlighting the multifactorial nature of emotional well-being.

The comparison across feature groups (Figure 33) illustrates these differences in performance, with academic context achieving the highest ROC-AUC, followed by sleep-related factors and the combined model. Although combining all features improves overall accuracy, this improvement is mainly driven by better identification of non-better days. The model still struggles to distinguish better-than-usual days, which is reflected in the more modest ROC-AUC values.

Overall, the results suggest that emotional variation is influenced by multiple interacting factors and cannot be easily captured using isolated behavioural signals. This reinforces both the complexity of the problem and the importance of considering personalised and context-dependent approaches.

Rather than indicating a lack of patterns, these findings suggest that emotional states are driven by highly individual and partially observable factors. While some general trends exist, their strength is limited and varies across users, making accurate prediction inherently difficult.

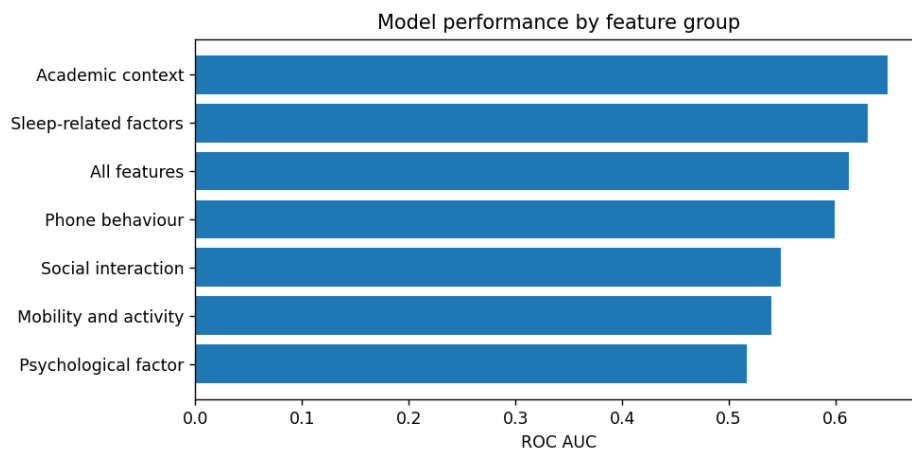


Figure 33. Model Performance (ROC-AUC) Across Feature Groups

8 DISCUSSION AND FINAL INSIGHTS

8.1 Overview of the Project

This project explores how daily emotional well-being can be analysed using data collected from multiple sources in real-life conditions. The work is based on the StudentLife dataset, which combines different types of behavioural, contextual, and self-reported data to analyse daily well-being.

One of the main challenges addressed in this project is the integration of different types of data. The dataset includes information collected at different frequencies and in different formats, including behavioural logs, smartphone sensing data, daily self-reported questionnaires, contextual and academic information, and survey-based measures. To deal with this, an initial data preparation stage was carried out, where all data sources were cleaned, aligned in time, and aggregated into a common daily structure. This resulted in a unified dataset where each observation represents one user on a specific day.

After building this analytical dataset, the project focuses on understanding emotional well-being from a daily and personalized perspective. The main outcome variable used throughout the analysis is daily happiness, obtained from ecological momentary assessment (EMA) reports. Since these reports are irregular and not always available, part of the work also considers the limitations of this type of data.

The analysis is developed progressively across several stages. First, an exploratory analysis is performed to understand the distribution and availability of the happiness variable, as well as to establish simple baseline models. These initial results show that predicting emotional states using general models is difficult, and that individual differences play a major role. In particular, personalized baselines provide better results, highlighting the importance of adopting a within-person perspective.

Based on this observation, the project adopts a within-person perspective, focusing on how emotional states vary over time for each individual. This leads to the analysis of emotional variability and stability, where both average happiness levels and day-to-day fluctuations are considered.

In parallel, survey-based psychological measures are used to define stable user profiles. These profiles provide a baseline understanding of individual differences and allow comparisons between groups of users with different well-being patterns.

Building on these analyses, the project introduces a similarity-based recommendation approach. Instead of relying on general patterns across users, this method compares each day with past days of the same individual to identify similar situations. Among these, only days associated with higher-than-usual happiness levels are considered, allowing the model to explore which contextual conditions are linked to improved well-being. In addition, a complementary rule-based analysis is introduced to identify more general behavioural patterns associated with better emotional states across each individual's history.

Finally, the project examines whether daily contextual and behavioural variables are related to better-than-usual emotional days. To do this, each day is classified as better or typical depending on whether

its happiness level is above the individual's average, and the analysis evaluates whether these variables can help distinguish between these two types of days.

On the whole, the project follows a consistent progression from data preparation to exploratory analysis, and from general observations to more personalized and context-aware methods. A key idea emerging from this work is that emotional well-being is highly individual, and that its analysis benefits from focusing on patterns within each person rather than relying only on global models.

8.2 Summary of Key Findings

The data preparation process highlighted the challenges of integrating multiple data sources and the importance of defining a consistent daily level of analysis to enable meaningful comparisons.

The exploratory analysis showed that daily happiness data were only available for a limited and irregular subset of the dataset, which represents an important constraint for the analysis. In addition, happiness values were not evenly distributed, with most observations concentrated at lower levels, while higher levels of happiness appeared less frequently. When simple predictive approaches were tested, the results indicated that general models had limited usefulness, as a linear regression model using contextual and behavioural variables performed similarly to a basic global average. However, when a personalized baseline based on each individual's average happiness was used, the prediction error decreased notably, highlighting the importance of individual differences and supporting the use of a within-person perspective. In addition, the comparison between better-than-usual and typical days suggested that emotionally better days are associated with different contextual and behavioural conditions.

Building on this, the analysis showed that emotional well-being cannot be fully described by average happiness alone, as individuals with similar mean levels can exhibit very different levels of day-to-day variability. No clear relationship was observed between average happiness and emotional variability, indicating that higher emotional levels do not necessarily imply greater stability. This highlights that emotional level and emotional stability represent distinct dimensions of well-being, and that considering both provides a more complete understanding of individual emotional patterns. In addition, substantial differences in variability were observed across users, with some individuals showing very stable patterns and others exhibiting strong day-to-day fluctuations.

The survey-based analysis showed that stable psychological measures provide a useful baseline for identifying meaningful differences between users. Using these measures, two coherent personality and well-being profiles were identified, one reflecting a more resilient and higher well-being pattern and the other a more vulnerable profile. These profiles were also reflected in daily life, as the more resilient group tended to show higher average happiness, more social interaction, and greater emotional stability, whereas the more vulnerable group showed lower happiness, greater emotional variability, and higher overall smartphone activity. All in all, these results show that personality and well-being measures from questionnaires provide a useful reference for understanding why people differ in their daily emotions and behaviours. However, these differences are moderate and show some overlap between profiles, indicating that personality profiles help explain, but do not fully determine, daily emotional and behavioural patterns.

The similarity-based analysis introduced a fully personalized approach to understanding emotional well-being, based on the idea that each individual can learn from their own past experiences. Instead of relying on general patterns across users, the method compares each day with similar past days of the same individual, identifying which contextual conditions tend to appear in days with higher happiness levels. The results showed that, by comparing the current day with similar past days where the user experienced higher-than-usual happiness, it is possible to gain insight into the contextual conditions associated with better-than-usual days, even when the current day is already relatively positive. Importantly, the method produces interpretable and situation-specific recommendations, allowing users to understand which aspects of their daily context are linked to higher happiness levels. To complement this perspective, a rule-based approach was also introduced, capturing more general patterns across the user's history. Together, both approaches provide a more complete and personalized understanding of well-being, combining insights from specific situations with broader behavioural trends.

The machine learning analysis examined whether daily contextual variables contain enough information to distinguish better-than-usual days from typical ones. The results showed that this is a challenging task, as all models achieved only moderate performance, indicating that better-than-usual days are difficult to identify based on contextual data alone. Some feature groups, particularly academic context and sleep-related factors, provided more useful information than others, although no single group was sufficient on its own. Combining multiple contextual variables did not lead to a clear improvement in model performance, as results remained moderate and no consistent gain was observed compared to individual feature groups. In addition, the models were consistently more effective at identifying typical days than better-than-usual days, suggesting that positive emotional deviations are harder to capture. The feature importance analysis further showed that no single variable dominates the predictions, with multiple factors contributing small amounts of information. Overall, these findings highlight the complexity of emotional well-being and suggest that better-than-usual emotional states cannot be easily identified using isolated behavioural signals. Moreover, these results also suggest that there are no strong universal behavioural patterns that consistently explain better-than-usual emotional states across users.

8.3 Interpretation and Discussion

Taken together, the results of this project suggest a clear idea: emotional well-being is highly individual, context-dependent, and difficult to fully capture using general models. Across the different analyses, it becomes evident that people do not experience their daily lives in the same way. This highlights the importance of looking beyond general patterns and focusing on the individual level.

The exploratory analysis already pointed in this direction, showing that general predictive approaches provide limited value, while personalized baselines offer a more meaningful way to interpret emotional states. This is reinforced by the analysis of emotional variability, which shows that individuals with similar average happiness can still experience very different day-to-day emotional patterns. In this sense, emotional well-being cannot be understood as a single stable value, but rather as a dynamic process that changes over time.

The survey-based analysis adds another layer to this interpretation. By introducing stable psychological measures, it becomes possible to better capture differences between individuals. The identification of distinct profiles shows that these profiles differ in how users typically feel and behave.

Building on these findings, the similarity-based approach provides a more intuitive and personalized way of analysing emotional well-being. Instead of trying to generalize across users, it focuses on each individual's own past experiences, allowing patterns to emerge in a more natural and interpretable way. This perspective suggests that people may benefit more from understanding their own behavioural patterns than from relying on general recommendations derived from others.

At the same time, the machine learning analysis highlights the limitations of relying solely on contextual variables to identify better-than-usual emotional days. The moderate performance of all models suggests that the selected variables do not fully capture the complexity of emotional states. The fact that models perform better at identifying typical days than positive deviations further reinforce the idea that emotionally better days are more subtle and harder to detect.

In summary, these results point towards a broader interpretation: emotional well-being is complex, influenced by many factors, and deeply personal. While behavioural and contextual data can help understand emotional states, they do not fully explain them. For this reason, approaches that focus on each individual and help them understand their own patterns may provide a more useful way to support well-being in everyday life.

8.4 Methodological Contributions

This project contributes to the analysis of emotional well-being by proposing a structured approach to analysing emotional well-being using real-world data from multiple sources. One of the main contributions lies in the integration of these data sources into a unified daily-level dataset, allowing behavioural, contextual, and self-reported information to be analysed in a consistent and comparable way.

A second key contribution is the adoption of a within-person analytical perspective. Instead of relying on comparisons across users, the analysis focuses on each individual. This approach allows emotional states to be interpreted in a more meaningful way, especially when strong individual differences are present. Building on this perspective, better-than-usual days are defined relative to each individual's average happiness, providing a simple and effective way to capture emotional variation within each user.

The project also introduces a similarity-based approach as a methodological contribution for analysing behavioural patterns. By comparing each day with similar past days of the same individual, this method provides a simple and understandable way to identify which daily conditions tend to be different in days with higher happiness levels. This approach differs from traditional predictive models by focusing on within-person comparisons and emphasizing interpretability and personalization.

In addition, the combination of different analytical perspectives, including exploratory analysis, profile-based grouping, similarity-based comparison, and machine learning models, provides a comprehensive

methodological framework. Rather than relying on a single technique, the project shows how multiple approaches can be integrated to analyse emotional well-being from complementary angles.

Finally, the use of machine learning models as an analytical tool, rather than as a purely predictive solution, represents another relevant methodological aspect. In this context, the models are used to explore the extent to which contextual variables contain detectable signals, contributing to a better understanding of the strengths and limitations of behavioural data for studying emotional well-being.

8.5 Limitations of the Study

One of the main limitations of this study is related to the availability and quality of the happiness data. Daily emotional states are measured using EMA reports, which are collected irregularly and are not available for all users and days. As a result, the analysis is based on a relatively small subset of observations, which may limit the robustness of the findings.

In addition, missing data are present across several variables, particularly in self-reported measures, which depend on user participation and are therefore not consistently recorded. Furthermore, many participants have limited data availability, with missing information across multiple days. This further reduces the number of complete observations available for some analyses, especially when multiple variables are considered simultaneously.

Another important limitation concerns the size of the analytical samples used in specific stages of the analysis. For example, some predictive models are trained on a reduced number of observations due to data availability constraints, and the variability analysis is based on a subset of users with sufficient data. As a result, the findings should be interpreted as exploratory rather than fully generalizable.

Additional limitations arise from the use of survey-based measures to define personality and well-being profiles. Although these variables provide a stable and interpretable baseline, they represent a static snapshot of the individual and do not capture possible changes over time. Furthermore, survey data rely on self-reported information, which may be affected by subjective bias and reporting inaccuracies. As a result, the identified profiles should be interpreted as general tendencies rather than precise or fixed representations of individuals.

The similarity-based approach also presents some methodological limitations. First, it relies on a relatively small number of historical observations for each user, which may limit the robustness of the recommendations. Furthermore, the method depends on the availability of past better-than-usual days, and in cases where no similar past days with higher happiness are identified, the system may be unable to generate meaningful recommendations. In addition, missing values are handled by replacing them with 0 after normalization, which corresponds to treating them as typical values for each user. This allows distances to be computed without discarding days, but it may reduce the model's ability to distinguish between different situations, as days with missing information may appear more similar than they really are.

Finally, the machine learning analysis is subject to several limitations related to both the data and the modelling approach. The dataset used for classification is relatively small and imbalanced, which makes

it difficult for the models to learn stable patterns. In addition, the definition of better-than-usual days as a relative outcome introduces ambiguity in the classification task, as similar contextual values may correspond to different labels depending on the individual. As a result, model performance remains moderate and the findings should be interpreted as indicative rather than conclusive.

8.6 Practical and Conceptual Implications

The results of this project have several practical implications for the design of digital well-being tools. In particular, the findings suggest that approaches focusing on individual patterns may be more useful than general recommendations, as individuals may respond differently to similar conditions [4]. Systems that analyse personal behavioural history and provide interpretable insights can help users better understand their own emotional dynamics and make more informed decisions in their daily lives. In this sense, the similarity-based approach illustrates how personal data can be transformed into meaningful and actionable feedback, supporting self-awareness rather than relying solely on automated predictions.

From a broader perspective, these results also have conceptual implications for the study of emotional well-being. The analysis shows that well-being cannot be understood as a fixed value, but rather as something that changes over time within each individual.

In addition, the findings indicate that behavioural and contextual data can help to understand daily happiness, but they do not fully explain it. This is reflected in the difficulty of identifying better-than-usual days using general models, as similar daily conditions do not always lead to higher happiness levels. This suggests that emotional states depend not only on what people do, but also on how each individual experiences those situations.

For this reason, analysing each person using their own data provides a more meaningful way to understand emotional well-being. By focusing on individual patterns, it becomes possible to identify what tends to be associated with higher happiness for each user, instead of relying only on general patterns that may not apply equally to everyone.

8.7 Future Work

Several directions can be explored to extend and improve the results of this project. First, future work could focus on using larger and more complete datasets, with more consistent emotional reports across users and time. This would allow more robust analyses and improve the ability to detect patterns associated with better-than-usual emotional states.

Future work could also explore alternative strategies for handling missing data, such as K-nearest neighbours (KNN) imputation. Instead of replacing missing values with zero or with the median, this method estimates missing values from similar observations based on the available variables. This could help preserve variability in the data and reduce the risk of treating missing information as typical behaviour.

Another line of work involves improving the representation of emotional well-being. Instead of relying on a single daily happiness value, future studies could incorporate multiple emotional dimensions, such as stress, sleep quality, or information about daily activities, providing a richer and more detailed understanding of daily emotional experience.

In addition, future research could analyse how emotional states change over time, instead of treating each day as independent. Considering sequences of days may help to better understand how past experiences influence current emotional states and identify patterns linked to improvements in happiness. In this context, sequence-based machine learning models, such as recurrent neural networks (e.g., LSTM), could be explored to explicitly capture temporal dependencies between consecutive days.

Another relevant direction is to further improve the personalized approach. The similarity-based method could be extended by giving different importance to the variables, so that the comparison between days focuses more on the factors that are most relevant for each individual. This could help identify more meaningful past situations and improve the quality of the recommendations.

Finally, an important extension would be the integration of these methods into real-world applications. Developing interactive systems that provide personalized feedback based on users' own data could help translate these analytical insights into practical tools for supporting emotional well-being in everyday life.

8.8 Final Conclusion

This project explored daily emotional well-being through the analysis of personalized data, with a particular focus on understanding how happiness varies across individuals in real-life conditions. The results suggest that well-being cannot be fully captured through general patterns, as individuals differ not only in their average levels of happiness, but also in their level of variability.

The survey-based analysis adds a broader perspective by identifying distinct user profiles, highlighting different patterns of behaviour and well-being.

At the same time, the results indicate that general predictive approaches based on behavioural and contextual data have limited ability to identify better-than-usual days. In contrast, approaches that focus on each individual's own data, such as the similarity-based method, offer a more interpretable and meaningful way to identify patterns associated with higher happiness levels.

As a whole, this work suggests that understanding daily well-being requires combining multiple perspectives, from general patterns to individual-level analysis. By integrating different approaches, it becomes possible to obtain a more complete and realistic view of how happiness is experienced in everyday life.

This reinforces the value of personalized data analysis as a key approach for understanding and supporting well-being in real-life contexts.

9 ETHICAL EVALUATION OF THE PROJECT

9.1 Introduction

This section presents the ethical perspective of the project. The aim is to reflect on the implications of using personal data related to behaviour and emotional well-being, even when this data comes from a publicly available research dataset.

The project analyses daily emotional well-being using the StudentLife dataset, which combines data from different sources. Because part of this information concerns mental health indicators such as depression, perceived stress or loneliness, an ethical reflection is particularly relevant here.

The reflection is divided into two parts. The first part covers the normative aspects, focused on the integrity of the research practices. The second part covers the transversal aspects, looking at the relationship between this kind of technology and society.

9.2 Normative Aspects

9.2.1 Integrity of the Research Practices

The project is based on the StudentLife dataset, originally collected at Dartmouth College and published as part of an academic publication. The data is reused under its original conditions: participants in the original study signed an informed consent form, and the study was approved by the Institutional Review Board at Dartmouth College [4]. No new data has been collected for the present project, no participant has been contacted, and no intervention has been carried out.

The dataset is properly cited, and every preprocessing step is documented in detail. Each chapter reports its actual sample sizes and explicitly states its limitations. The conclusions are framed as exploratory rather than as universal findings, which is a deliberate decision to avoid presenting moderate results as if they were stronger than they really are.

9.2.2 Privacy and Data Sensitivity

The data used in this project is sensitive, since variables derived from the PHQ-9, the Perceived Stress Scale, the loneliness scale and EMA mood reports relate to mental health. This kind of information always requires a careful approach, even when the dataset has already been anonymised.

Anonymisation does not completely eliminate the risk of identification, especially in a small sample of around 49 participants from a specific course in 2013. For this reason, the dataset is used only for academic analysis within the scope of this project and is not shared or redistributed.

9.2.3 Privacy-Conscious Design Decisions

Several decisions during the data preparation stage were taken to keep the analysis focused on less intrusive variables:

- The GPS, WiFi location and conversation sensing modules were excluded from the analysis. These signals were considered too intrusive or unnecessary for the objectives of the work.
- The free-text EMA responses (the 'comment' and 'describe' modules) are used only as participation indicators. The number of non-empty responses is counted, but no semantic or sentiment analysis is performed on the actual text, in order to avoid both unnecessary complexity and additional privacy concerns.

Other variables already came in a privacy-preserving form from the original dataset [4]: the audio module provides only categorical labels (silence, voice, noise) instead of recordings, and call and SMS data are reduced to hashed event counts. The project simply uses these variables in the form in which they were released.

Together, these decisions reflect a privacy-conscious approach that keeps only the variables that are needed for the analysis.

9.3 Transversal Aspects

9.3.1 Actors Involved

Several actors are involved, directly or indirectly, in the ethical dimension of this project:

- The participants of the original StudentLife study, whose information is ultimately being analysed.
- The authors of the original study and the institution where it was conducted, who designed the data collection protocol, obtained informed consent and made the dataset available for further research.
- The student carrying out this project, Adriana Rodriguez, who is responsible for taking ethical decisions during data preparation, analysis and how the results are communicated.
- The academic supervisor and the university, who provide guidance and ensure that the work follows the ethical standards expected from a final degree project.
- Potential future users, in case the methods developed here were ever applied in a real-world setting.

Listing these actors helps to show that ethical responsibility is shared: some decisions came from the original study, others belong to this project, and others would fall on anyone who applied these ideas in a real setting.

9.3.2 Values and Principles

Four ethical principles guide this work: autonomy, non-maleficence, beneficence and justice.

Autonomy refers to the right of individuals to make decisions about their own data. This principle is respected at the level of the original consent, and also in the design of the recommendation approach, which presents patterns from the user's own past experiences for reflection rather than telling them what to do.

Non-maleficence is the principle of avoiding harm. The analyses are descriptive and exploratory, the methods are not intended for clinical assessment or diagnosis, and the most intrusive variables were deliberately excluded.

Beneficence means aiming to do good through the work. The contribution here is modest: this project does not directly improve anyone's well-being, but it shows how data about people can be analysed in a responsible, interpretable and person-centred way, which can be useful for future research on similar topics.

Justice means being fair about who the conclusions apply to. Since the dataset comes from a single university in 2013 with a very specific student population, presenting the findings as if they described students or young people in general would be unfair to anyone whose situation is different.

9.3.3 Risks and Social Impact

Working with this kind of data involves several risks.

The first risk is misinterpreting what the system actually produces. The project combines passive sensing data, behavioural and academic information with self-reported answers and questionnaires to detect patterns, classify days and generate personalised recommendations, but none of these outputs can really capture how a person feels inside. Emotions depend on many personal factors that no model can fully describe, so the results should be seen as exploratory patterns, never as objective measures of someone's emotional state.

The second risk is that part of the data depends on how participants felt at the moment of answering (EMA data). Self-reported answers about mood, stress or happiness are subjective, can change quickly, and are influenced by many things outside the project's control, which adds uncertainty to the results.

The third risk is the possible emotional impact on the user. Receiving feedback about one's own mood patterns can be helpful for some people, but for others it could increase self-criticism or anxiety, especially during difficult periods. Any tool of this kind should consider that not every user will react in the same way.

Acknowledging these risks is part of using the project's results responsibly.

9.3.4 Actions to Prevent Negative Impacts

Several concrete actions have been taken to address the risks discussed above:

- The project is kept within an academic and exploratory scope, framing all results as patterns rather than as objective measures of someone's emotional state.
- The number of participants and observations used in each analysis is reported clearly, so the results are not interpreted as stronger than they really are.
- The most intrusive variables (GPS, Wi-Fi location, free-text content) were excluded from the analysis, and the dataset is kept only for academic use within this project.

- The recommendation system is presented as a reflective tool, designed to support self-awareness without telling the user how to feel or what to do.
- The methods are not meant to be used as diagnostic or prescriptive tools, and they should not replace professional mental health support if this kind of approach were ever applied with real users.

9.4 Conclusion

The ethical perspective of this project is shaped by the fact that it works with sensitive information, even though the data has been collected by others, anonymised and made publicly available.

The main ethical strengths of the work are the within-person and interpretable approach, the deliberate exclusion of the most intrusive variables, and the framing of the recommendation system as a reflective tool rather than a diagnostic one. The main limits come from the data itself: a small and specific sample and the difficulty of fully capturing emotional states from behavioural signals.

Within these limits, the project tries to balance the analytical value of the data with the responsibility that comes from working with information about people's daily lives and mental health.

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ACRONYMS

Table 17. Acronyms

Acronym	Meaning
BFI-44	Big Five Inventory (44 items)
EMA	Ecological Momentary Assessment
GPA	Grade Point Average
GPS	Global Positioning System
IRB	Institutional Review Board
KNN	K-Nearest Neighbours
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
NaN	Not a Number
PAM	Photographic Affect Meter
PANAS	Positive and Negative Affect Schedule
PCA	Principal Component Analysis
PHQ-9	Patient Health Questionnaire-9
PSQI	Pittsburgh Sleep Quality Index
PSS-10	Perceived Stress Scale (10 items)
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SD	Standard Deviation
SMS	Short Message Service
UCLA	University of California, Los Angeles
UTC	Coordinated Universal Time
VR-12	Veterans RAND 12-Item Health Survey
Wi-Fi	Wireless Fidelity

APPENDIX A

This appendix provides a detailed description of all variables used in the analysis, including their source and definition.

Table A1. Identifiers

Variable	Block	Source	Description
user_id	Identifiers	Integrated dataset	Unique participant identifier
date	Identifiers	Integrated dataset	Daily observation date

Table A2. Behavioural Measures

Variable	Block	Source	Description
total_apps_used	Behavioural measures	app_usage	Number of distinct apps used per day
most_used_app	Behavioural measures	app_usage	Most frequently used app
num_sessions	Behavioural measures	app_usage	Number of usage sessions
tiempo_uso_total	Behavioural measures	app_usage	Total app usage activity
calendar_event_count	Behavioural measures	calendar	Number of calendar events
total_calls	Behavioural measures	call_log	Total number of calls
total_meals	Behavioural measures	dinning	Total meals recorded
breakfast_events	Behavioural measures	dinning	Breakfast occurrences
lunch_events	Behavioural measures	dinning	Lunch occurrences
supper_events	Behavioural measures	dinning	Dinner occurrences

Table A3. Academic and Educational Measures

Variable	Block	Source	Description
num_classes	Academic measures	class.csv	Number of enrolled classes
grade_mean	Academic measures	grades.csv	Mean academic performance
grade_std	Academic measures	grades.csv	Variability in grades
num_deadlines_today	Academic measures	deadlines.csv	Number of deadlines
num_posts	Academic measures	piazza.csv	Posts in forum
num_responses	Academic measures	piazza.csv	Responses in forum
piazza_score	Academic measures	piazza.csv	Participation score

Table A4. EMA Emotional Measures

Variable	Block	Source	Description
happy	EMA emotional	Mood	Daily happiness
sad	EMA emotional	Mood	Daily sadness
tomorrow	EMA emotional	Mood	Expected next-day mood
how	EMA emotional	Mood	Current emotional state
stress_level	EMA emotional	Stress	Daily stress level
positive_mean	EMA emotional	Events	Positive event intensity
negative_mean	EMA emotional	Events	Negative event intensity
anxious	EMA emotional	Behaviour	Anxiety level
calm	EMA emotional	Behaviour	Calmness level
enthusiastic	EMA emotional	Behaviour	Enthusiasm
sympathetic	EMA emotional	Behaviour	Sympathy
disorganized	EMA emotional	Behaviour	Disorganization
reserved	EMA emotional	Behaviour	Reserved behaviour
dependable	EMA emotional	Behaviour	Dependability

Table A5. EMA Contextual Measures

Variable	Block	Source	Description
working	EMA contextual	Activity	Working activity level
relaxing	EMA contextual	Activity	Relaxing activity level
other_working	EMA contextual	Activity	Other working-related activity
other_relaxing	EMA contextual	Activity	Other relaxing-related activity
Social2	EMA contextual	Activity	Social activity indicator
experience	EMA contextual	Class	Class experience
hours	EMA contextual	Class	Study hours
due	EMA contextual	Class	Perceived workload
challenge	EMA contextual	Class 2	Task difficulty
effort	EMA contextual	Class 2	Effort level
grade	EMA contextual	Class 2	Expected grade
sleep_hours	EMA contextual	Sleep	Sleep duration
sleep_quality	EMA contextual	Sleep	Sleep quality
social_people	EMA contextual	Social	Social exposure
study_productivity	EMA contextual	Study Spaces	Productivity
study_place	EMA contextual	Study Spaces	Study location
breakfast_place	EMA contextual	Dining Halls	Breakfast location
lunch_place	EMA contextual	Dining Halls	Lunch location
dinner_place	EMA contextual	Dining Halls	Dinner location

Table A6. EMA Engagement Measures

Variable	Block	Source	Description
comment_count	EMA engagement	Comment	Number of responses
dimensions_count	EMA engagement	Dimensions	Number of EMA answers

Table A7. Passive Sensing Measures

Variable	Block	Source	Description
prop_stationary	Passive sensing	activity inference	Time stationary
prop_active	Passive sensing	activity inference	Time active
prop_voice	Passive sensing	audio inference	Voice presence
prop_noise	Passive sensing	audio inference	Noise presence
bt_unique_devices	Passive sensing	bluetooth	Unique devices
bt_scans	Passive sensing	bluetooth	Scan frequency
dark_minutes	Passive sensing	dark	Screen inactivity
charge_minutes	Passive sensing	phonecharge	Charging time
locked_minutes	Passive sensing	phonelock	Locked time
unique_wifi	Passive sensing	wifi	WiFi diversity
num_sms	Passive sensing	sms	SMS activity

Table A8. Survey-based Psychological Traits

Variable	Block	Source	Description
extraversion	Survey traits	BigFive	Extraversion
agreeableness	Survey traits	BigFive	Agreeableness
conscientiousness	Survey traits	BigFive	Conscientiousness
neuroticism	Survey traits	BigFive	Neuroticism
openness	Survey traits	BigFive	Openness
flourishing_score	Survey traits	Flourishing	Well-being
loneliness_score	Survey traits	Loneliness	Loneliness
panas_positive	Survey traits	PANAS	Positive affect
panas_negative	Survey traits	PANAS	Negative affect
stress_score	Survey traits	PSS	Perceived stress
phq_score	Survey traits	PHQ-9	Depression score

APPENDIX B

Declaration on the Use of Artificial Intelligence Tools

This appendix declares the Artificial Intelligence (AI) tools used during the development of this project, the stages in which they were used, and the purpose of their use.

AI Tools Used

- ChatGPT (OpenAI)
- Claude (Anthropic)

Stages of Use and Purpose

The use of these tools was limited to a supporting role across several stages of the project:

- *Literature review.* AI tools were used to help summarize and clarify complex sections of scientific papers related to emotional well-being, personalized data analysis, and the StudentLife dataset. All sources were subsequently verified and read in their original form.
- *English writing.* As the memory is written in English, AI tools were used to improve grammar, clarity, and academic style in certain sections, and to suggest alternative phrasings. The content was always defined beforehand.
- *Support with Python code.* The methodological approach, the design of each analysis, and the logic of every method were thought out and defined by me. AI tools were consulted as a support when implementing specific parts in code, helping with particular methods, resolving syntax doubts, debugging errors, or suggesting alternative ways of writing a specific operation.
- *Structure and visualization ideas.* AI tools were consulted to help organize the structure of certain chapters and to brainstorm ideas on how to visually present some of the results.

Final Remarks

The use of AI was limited to a supporting role and did not replace the analytical, methodological, or interpretative work carried out throughout this project. All outputs generated with AI assistance were reviewed and validated personally. The methodology, analytical decisions, interpretation of results, ethical evaluation, and final conclusions are entirely my own work.